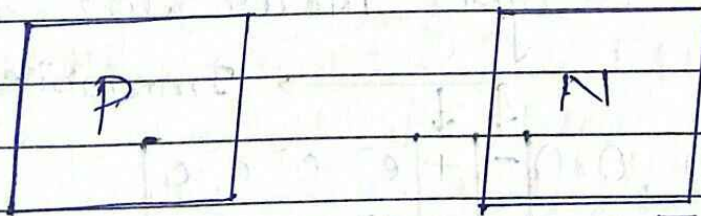


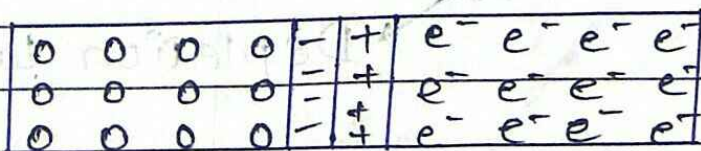
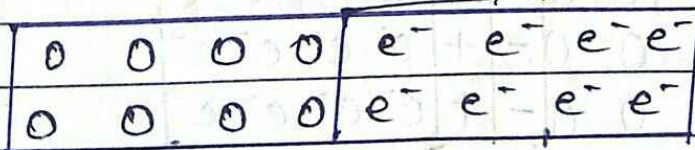
MODULE - 02

SEMICONDUCTOR DIODE

- 1) Formation of P-n Junction
- 2) Position of Fermi level in Equilibrium
- 3) V-I characteristics in Forward and Reverse bias.
- 4) Capacitance of P-n Junction diode.
- 5) Zener Diode
- 6) Zener Diode as a Voltage Regulator
- 7) Applications of Special Purpose diode viz. PIN diode, Schottky diode, Gunn diode, LED laser diode, Photo diode, Tunnel diode and Solar cell.
- 8) Diode Circuits:- Clipping, Clamping, Voltage multiplier and Rectifier.



→ P-N Junction.



* P-N Junction —

→ When Acceptor impurity added to one side and donor impurities added to other side of Silicon plate the P-N Junction is formed.

→ P-N Junction

0	0	0	0		e ⁻	e ⁻	e ⁻	e ⁻
0	0	0	0		e ⁻	e ⁻	e ⁻	e ⁻
0	0	0	0		e ⁻	e ⁻	e ⁻	e ⁻

* FORMATION OF DEPLETION LAYER/REGION.

When the Physical Contact between P-Type & N-Type semiconductor is made free e⁻ in N-Type diffuses across the junction into P-Type and diffusion of each e⁻ from N-Type leave a positive charge in the N-Type.

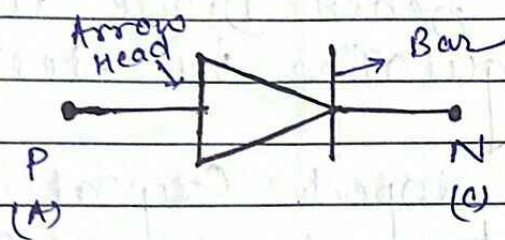
Near the, Similarly diffusion of each hole from P-Type leave a negative charge in P-Type near the Junction.

→ Immobile Ion

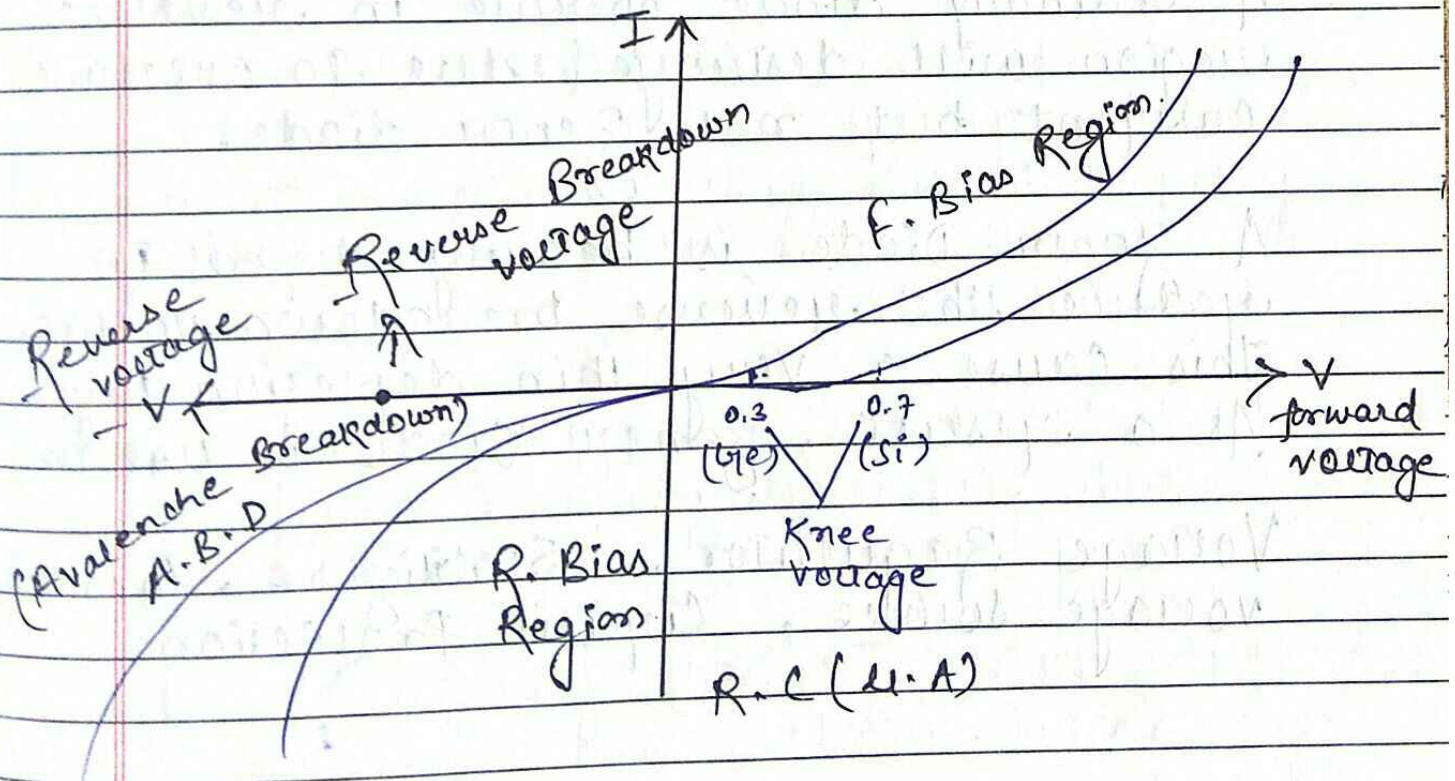
0	0	0	0	-	+	e ⁻	e ⁻	e ⁻	e ⁻
0	0	0	0	-	+	e ⁻	e ⁻	e ⁻	e ⁻
0	0	0	0	-	+	e ⁻	e ⁻	e ⁻	e ⁻
0	0	0	0	-	+	e ⁻	e ⁻	e ⁻	e ⁻

← w →
Depletion layer

Diode - A Two Terminal or Two electrode Semiconductor device which allow the electric current to flow only in one direction. While blocking the current in the opposite or reverse direction is known as diode or P-N Junction diode.



V-I characteristics of Diode :-



Zener Diode - Zener Diode also called Breakdown diode. It is a specially doped semiconductor device designed to operate in the reverse breakdown region.

It maintained constant voltage across load. This makes Zener Diode useful for voltage regulation in electronic circuit.

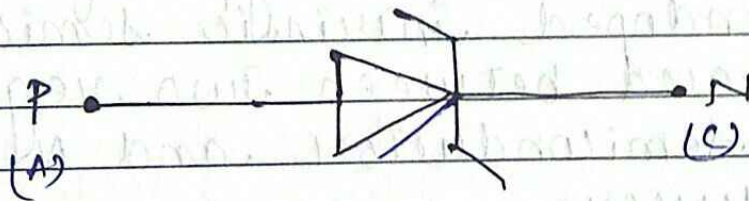
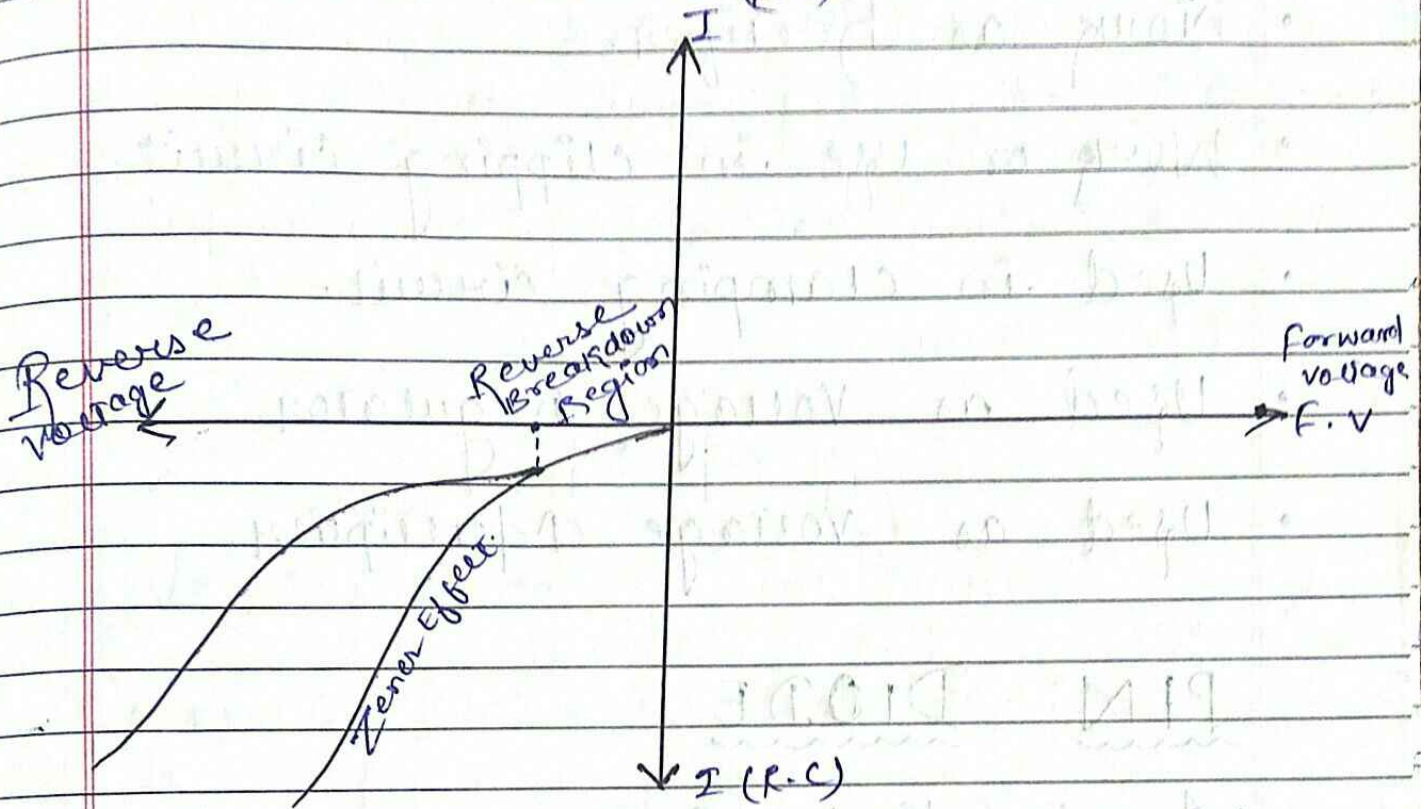
A properly doped Crystal Diode which has sharp breakdown voltage is known Zener diode.

A ordinary diode operate in reverse region will destroyed due to excessive current but not Zener diode.

A Zener Diode is heavily doped to reduce the reverse breakdown voltage. This cause a very thin depletion layer. As a result, Zener Diode is use as-

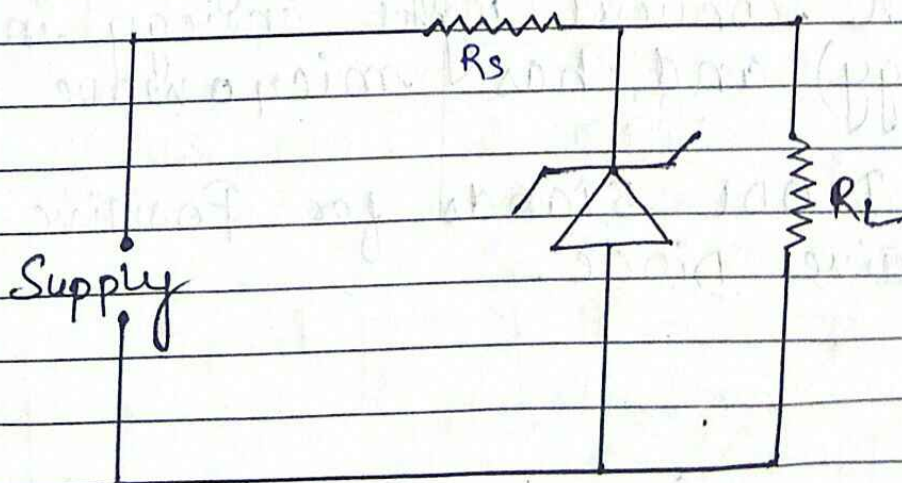
Voltage Regulator, Stabilizer, Constant voltage source, Circuit Protection.

V.V.D Zener Diode as a Voltage Regulator (P.C)



Zener Diode

or



Application of Special Purpose Diode:-

- Work as Rectifier.
- Work or use in clipping circuit.
- Used in clamping circuit.
- Used as Voltage Regulator.
- Used as Voltage Multiplier.

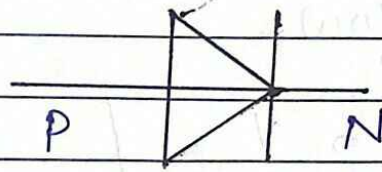
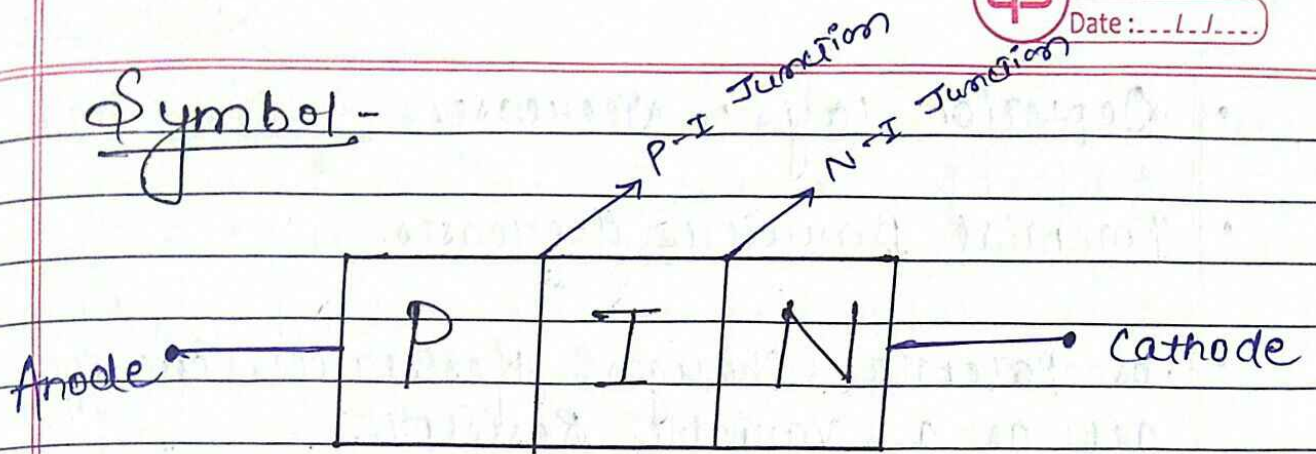
PIN DIODE

A pin diode is a type of diode having undoped intrinsic semiconductor region placed between two region (doped P-type semiconductor and N-type semiconductor).

PIN DIODE is a type of Photo detector (which convert light Energy into electrical Energy) and has microwave application.

PIN DIODE stands for Positive-Intrinsic-Negative Diode.

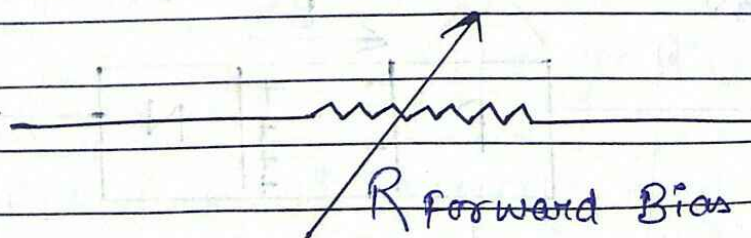
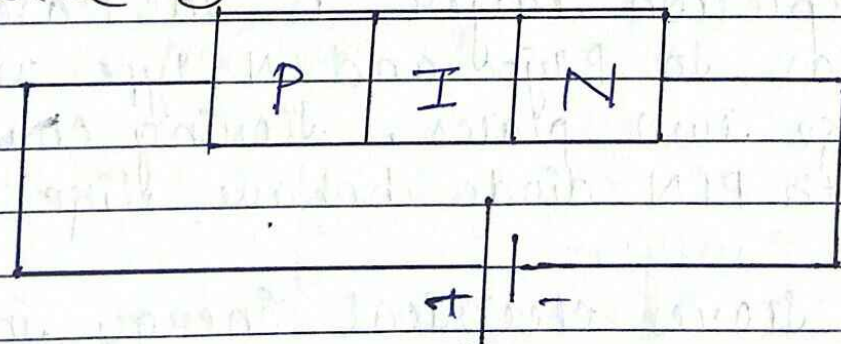
Symbol -



Working Mode of PIN DIODE

- 1) Forward Bias (As a Variable Resistor) ($\rightarrow$)
- 2) Reverse Bias (As a Capacitor) ($\rightarrow$)

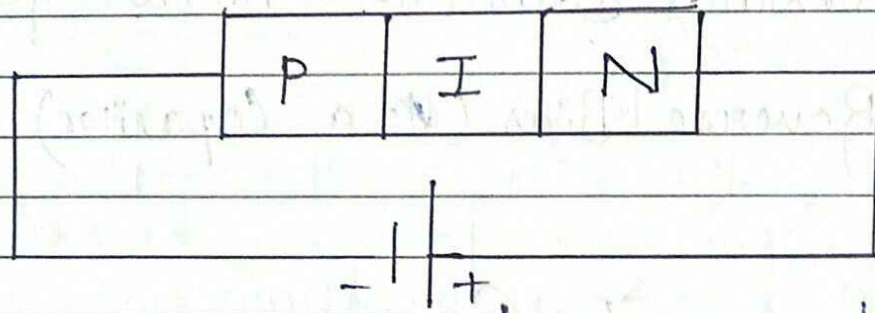
1) Forward Bias -



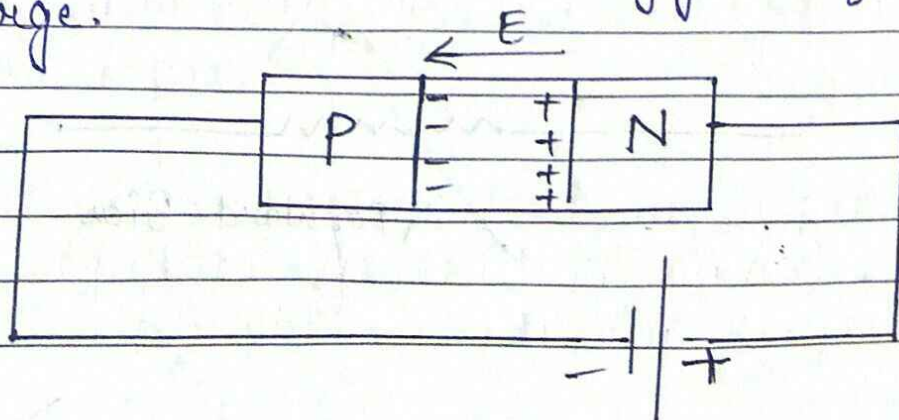
- Depletion layer decreases.
- Potential Barrier decreases.
- As Potential changes, Resistance change so act as a variable resistor.
- More Current flow;

$$I \uparrow \Rightarrow \frac{V}{R \downarrow}$$

27 Reverse Bias - (Capacitor) -



- Depletion layer is increased in Reverse Bias so P-type and N-type will behave like two plates, storing charge between both PIN diode behave like capacitor.
- It stores electrical Energy in form of charge.



Advantages -

- 1) PIN diode have low Noise.
- 2) It has very low bias voltage.
- 3) High capacitance
- 4) High Reverse breakdown voltage.

Disadvantages -

- 1) High Reverse Recovery Time

Application -

- 1) It used as High voltage Rectifier.
- 2) It used as Photo detector.
- 3) Used in Photovoltaic cell.
- 4) Used in low voltage Amplification.

GUNN DIODE

Gunn Diode also known as "TED".

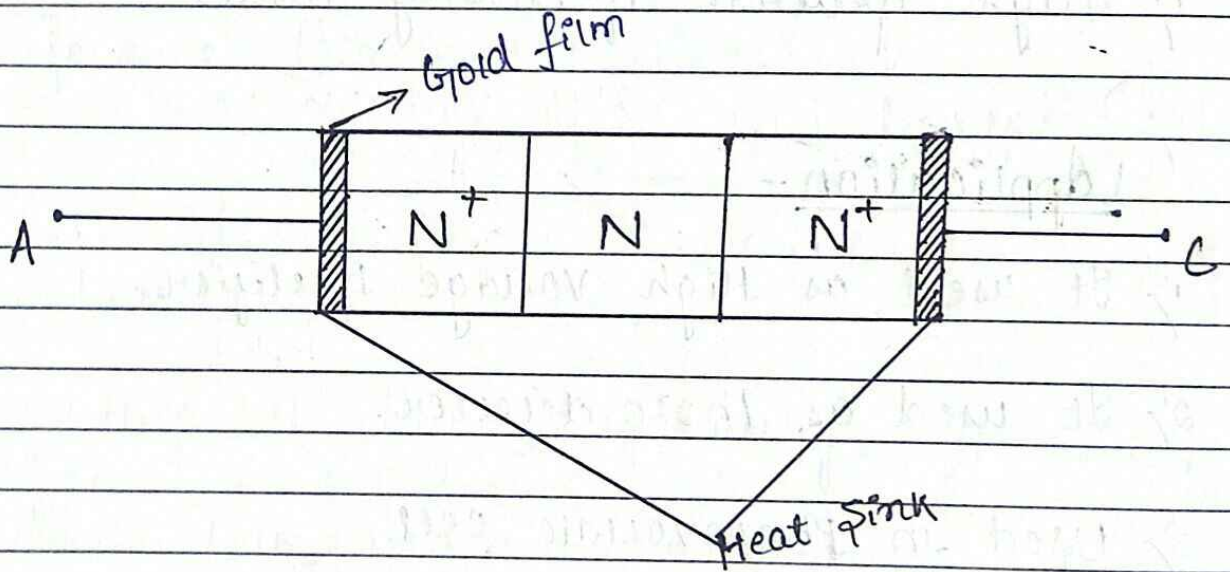
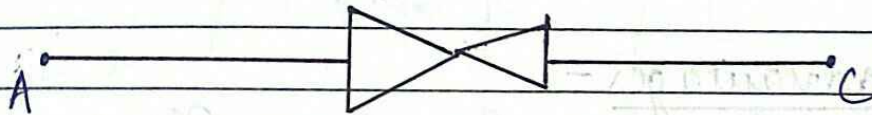
TED - Transferred electronic device.

Device which is composed of only one type of semiconductor.

It is used to generate Radiofrequency and Microwave frequency.

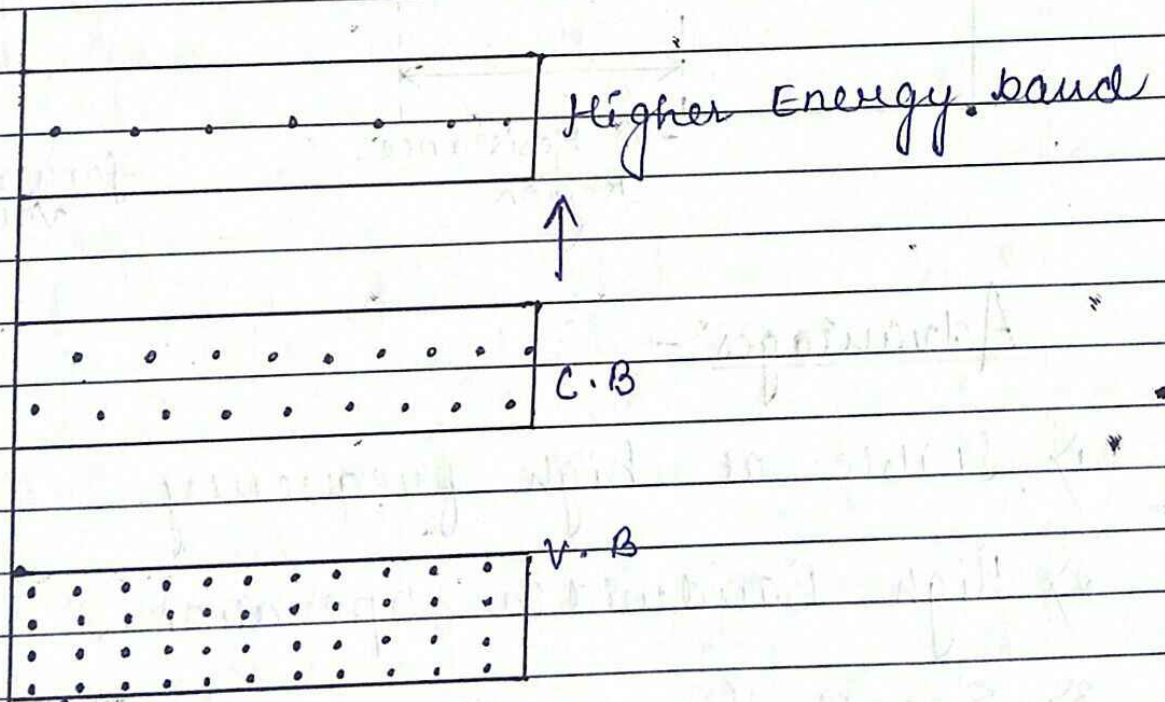
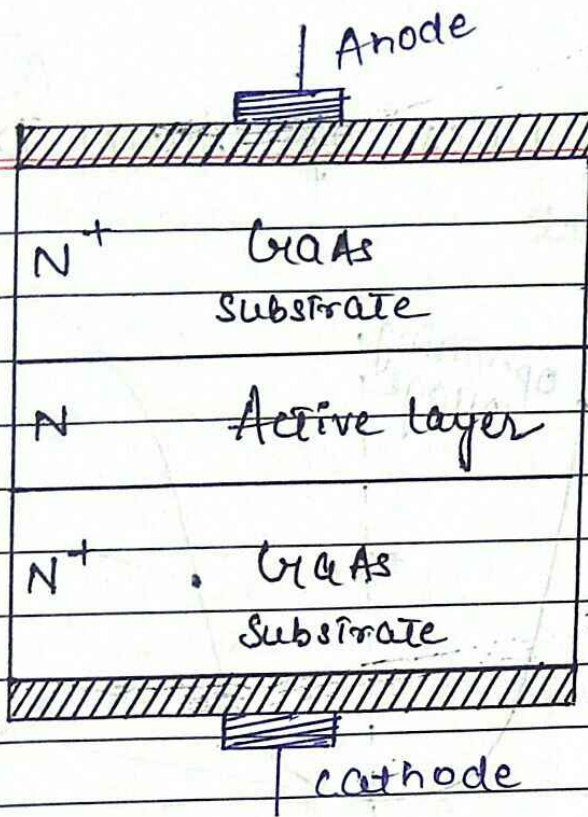
It is composed of only N-Type semiconductor because N-Type S.C has electron as majority carrier.

Symbol -



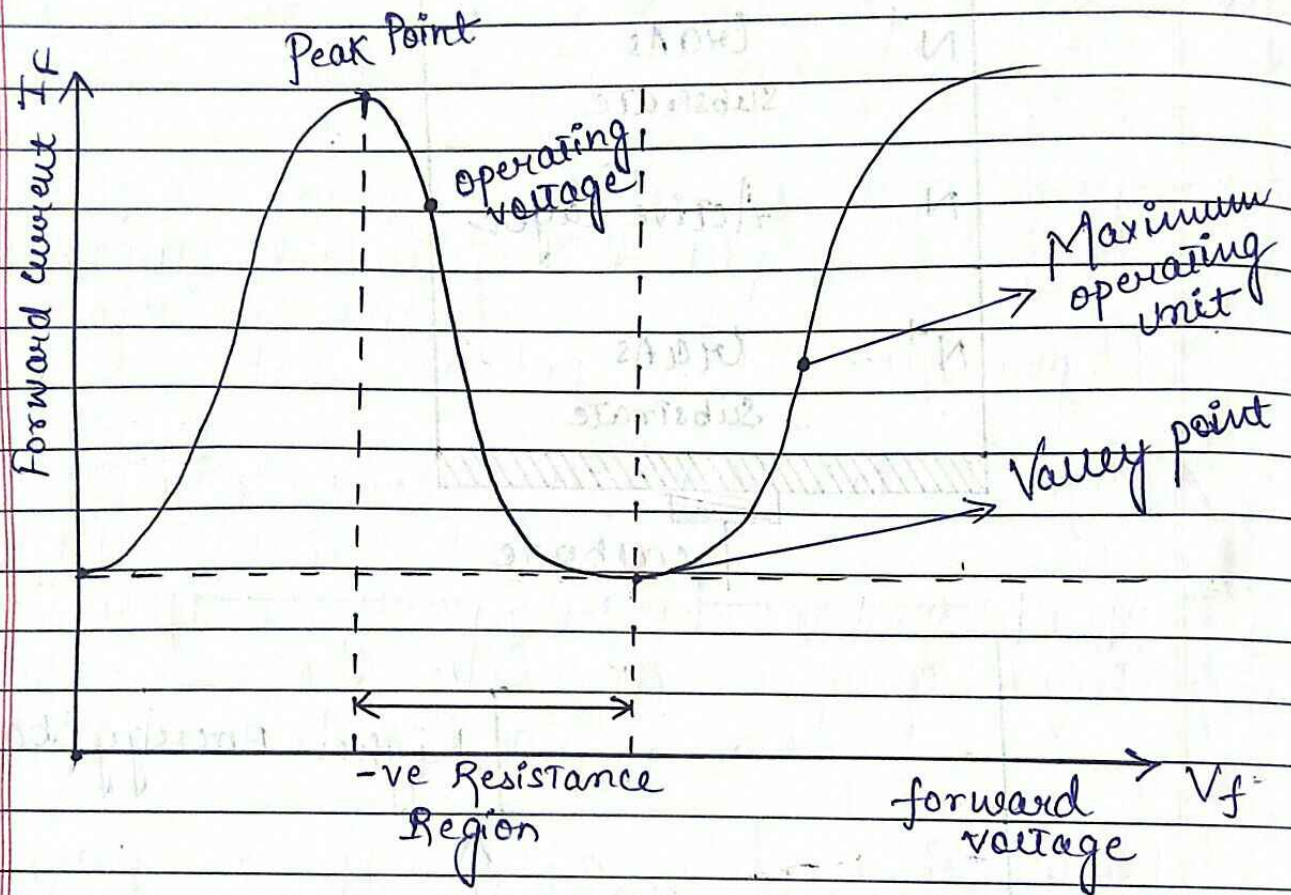
Construction :-

- 1) It is made up of three layer of N-type semiconductor.
- 2) The semiconductor used in Gunn diode are Gallium Arsenide (GaAs), Gallium Nitride (GaN), Cadmium Sulphide (CdS).



Among these 3 layers, the top & bottom layers heavily doped while the Middle layer is lightly doped.

V-I characteristics of GUNN DIODE



Advantages -

- 1) Stable at high frequency.
- 2) High Bandwidth operation.
- 3) Small Size & Portable.
- 4) Low Manufacturing Cost.

Disadvantages -

- 1) Poor Temperature Stability
- 2) Low Efficiency

Application -

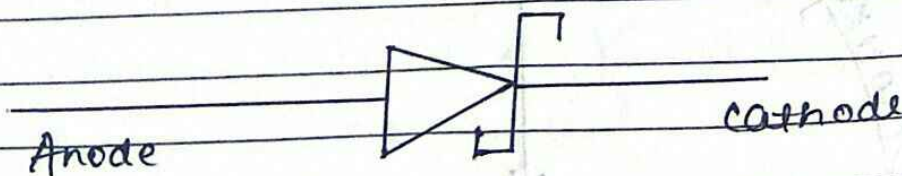
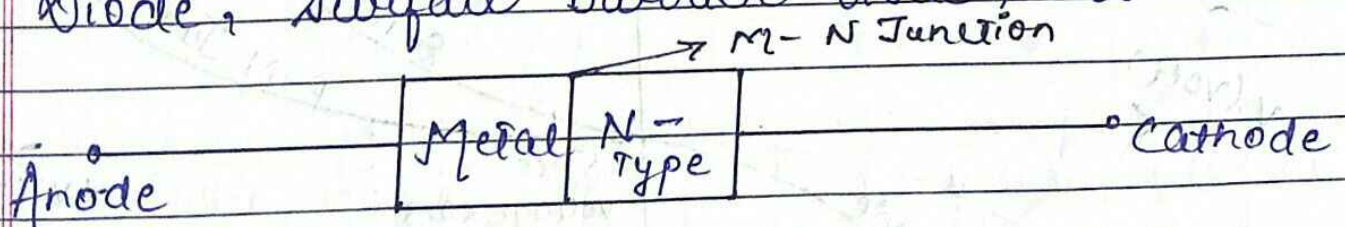
- 1) Used as oscillator & Amplifier.
- 2) Used in Radio-communication.
- 3) Used in fast controlling Equipment.
- 4) Used as sensor.

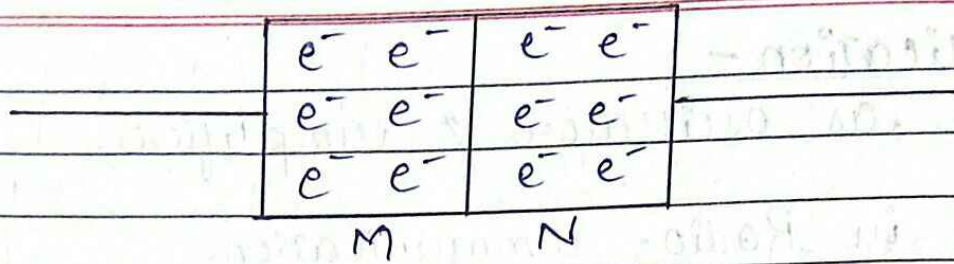
Schottky Diode

Schottky Diode is a Metal Semiconductor diode that has less forward voltage drop than P-N Junction diode and can be used in high speed switching Application.

In Schottky Diode, Metal such as Aluminium and Platinum replace the P-type semiconductor.

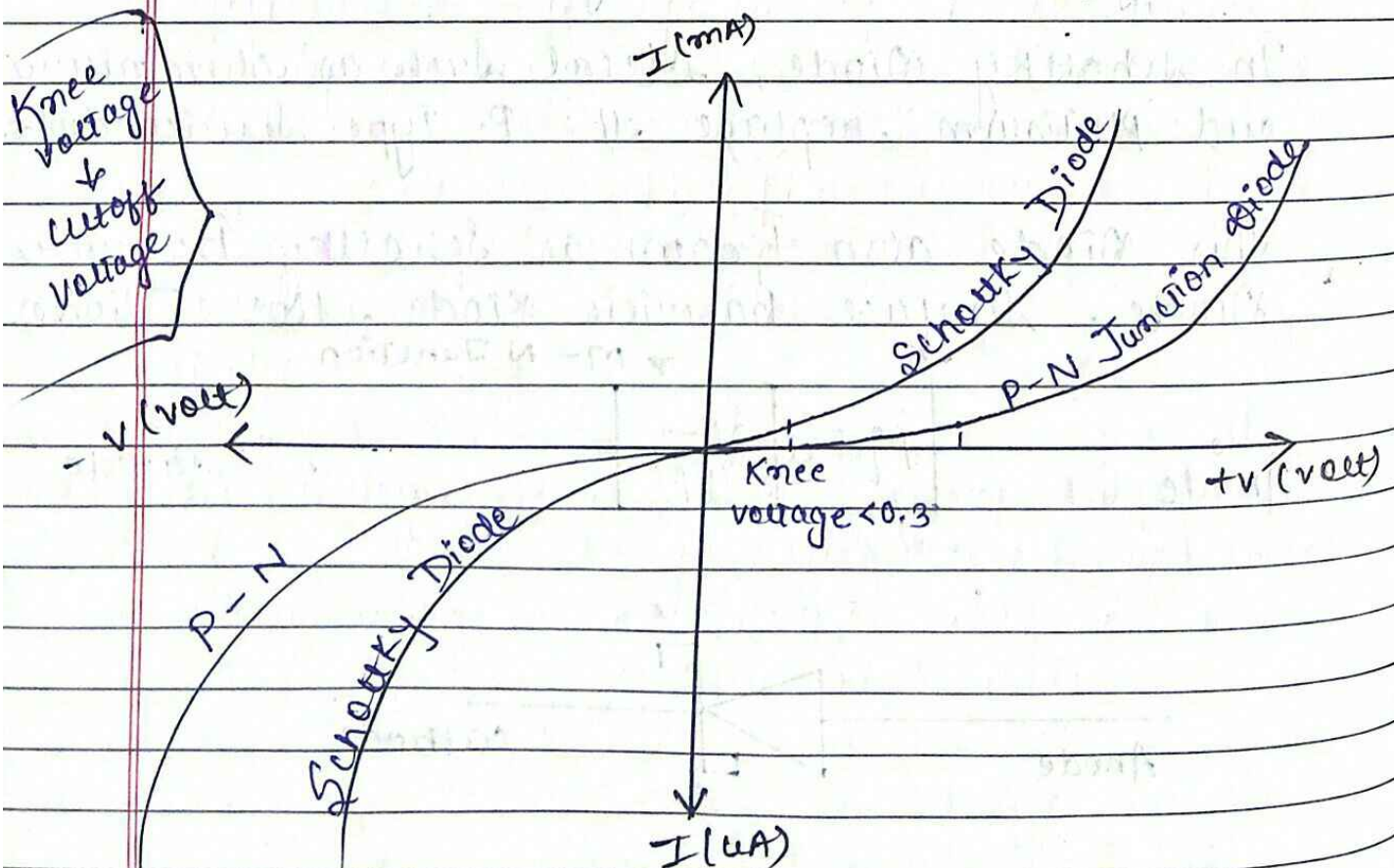
This Diode also known as Schottky Barrier Diode, Surface Barrier Diode, Hot e^- diode.





- 1) Very thin Depletion layer.
- 2) Across Depletion layer. E is Negligible
- 3) Potential Barrier less (0.3 eV).
- 4) Quick switching Action at high frequency.
- 5) Unipolar Device.

V-I Characteristics of Schottky Diode



Advantages -

- 1) low turn on voltage.
- 2) low forward drop.
- 3) fast Recovery Time.
- 4) High Efficiency
- 5) Operate at High frequency.

Disadvantages -

- 1) This is very sensitive device.

Application -

- 1) Used in voltage clamping circuit.
- 2) Used in Rectifier
- 3) Used in SMPS (switch Mode Power Supply).
- 4) Used in logic gate.
- 5) Used in Radio frequency.

Optical Diode

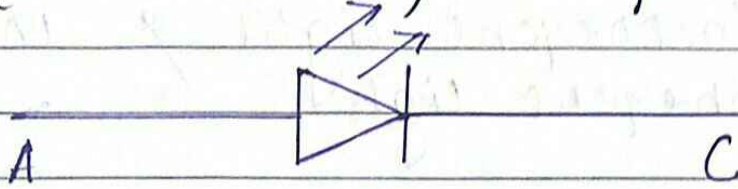
- 1) LED
- 2) LASER
- 3) PHOTO DIODE

1) LED (Light Emitting Diode) - It works on the principle of Electro-luminescence. LED is a diode that gives visible light when it is forward bias.

When charge carrier recombine i.e. electron recombine with hole then energy is emitted in the form of heat when silicon or germanium is used as semiconductor. But in case of other semiconductor material like Gallium, Phosphorus and Arsenic, a greater Percentage of Energy is released during Recombination in the form of light.

By Varying the Quantity of these element it is possible to produce light of different Wavelength with colour that include Red, Green, Yellow and Blue.

$\text{Ga} - \text{As} \Rightarrow \text{Infrared light}$
 $\text{Ga} - \text{P} \Rightarrow \text{Red or Green}$
 $\text{Ga} - \text{As} - \text{P} \Rightarrow \text{Red or Yellow}$



Application -

- 1) It can be used as a obstacle detector.
- 2) It can be Used as a Vigital counter.
- 3) It is used in Infrared camera.
- 4) Used in Remote control.

LASER DIODE

Laser Diode stands for light Amplification by stimulated Emission of Radiation. Laser is a device that emit light through a process of Optical Amplification based on the Stimulated emission of Electro-Magnetic Radiation. A laser diode convert electrical Energy into light Energy to produce high intensity of coherent light.

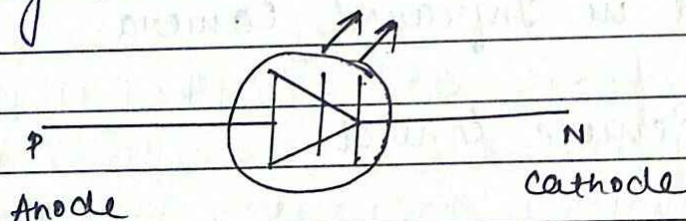
In a laser Diode, the P-N Junction of S.C Diode act as a laser Medium.

coherent
↓
{ phase diff.
between
their
wave
is
constant }

The Working of Laser Diode is almost similar to LED. The Main diff. between LASER & LED is LED emits incoherent light & LASER DIODE emit coherent light.

The laser diode is made up of two doped gallium arsenide (GaAs).

LASER DIODE is same as LED except the channel used in laser are narrow to produce a single beam of light.



Application of LASER DIODE -

- 1) Used in Laser Printer.
- 2) Used in fiber optic communication.
- 3) Used in Laser Pointer.
- 4) Used as Bar code Reader.
- 5) Used in Laser Scanner.

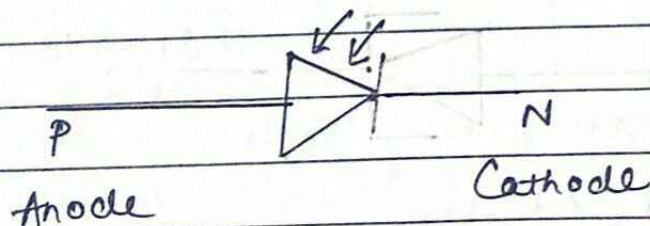
PHOTO-DIODE

A Photo Diode is a Reversed-Biased Silicon or Germanium P-N Junction which reverse current increased when the junction is exposed to light. The reverse current in a photo diode is directly proportional to the intensity of light falling on its P-N Junction.

Working Principle -

When light (photon) falls on the P-N Junction. The energy is imparted by the photon to the atom in P-N Junction. This will create more free e^- . The additional free e^- will increase the reverse current.

The intensity of incident light on P-N Junction increases, the reverse current also increases. On the other word, As the incident light increases, the resistance of the Photo Diode decreases.



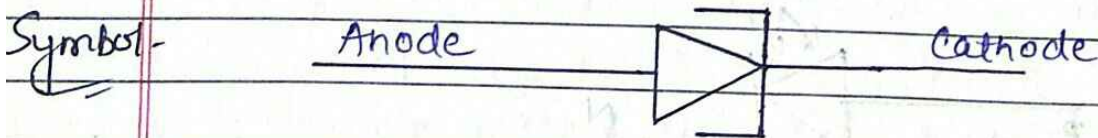
Application -

- 1) Used in Alarm circuit.
- 2) Used in Counter circuit.
- 3) Used in Optical communication.
- 4) Used in Switching circuit.

ASSIGNMENT - 1 - TUNNEL DIODE . 2 - SOLAR CELL .

TUNNEL DIODE - A Tunnel diode is a heavily doped P-n junction diode in which the electric current decreases as the voltage increases.

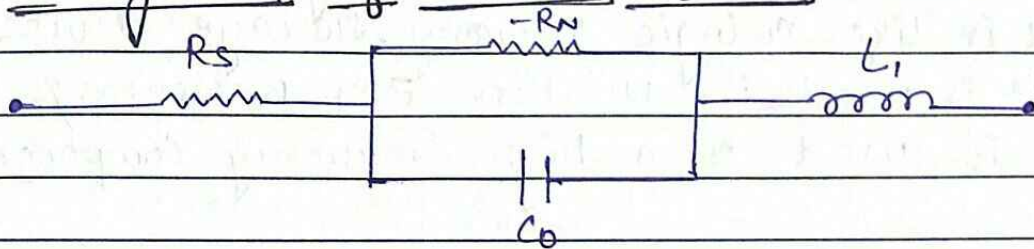
In Tunnel diode, electric current is caused by "Tunneling". The Tunnel diode is used as a very fast switching device in computers. It is also used in high-frequency oscillator and amplifiers.



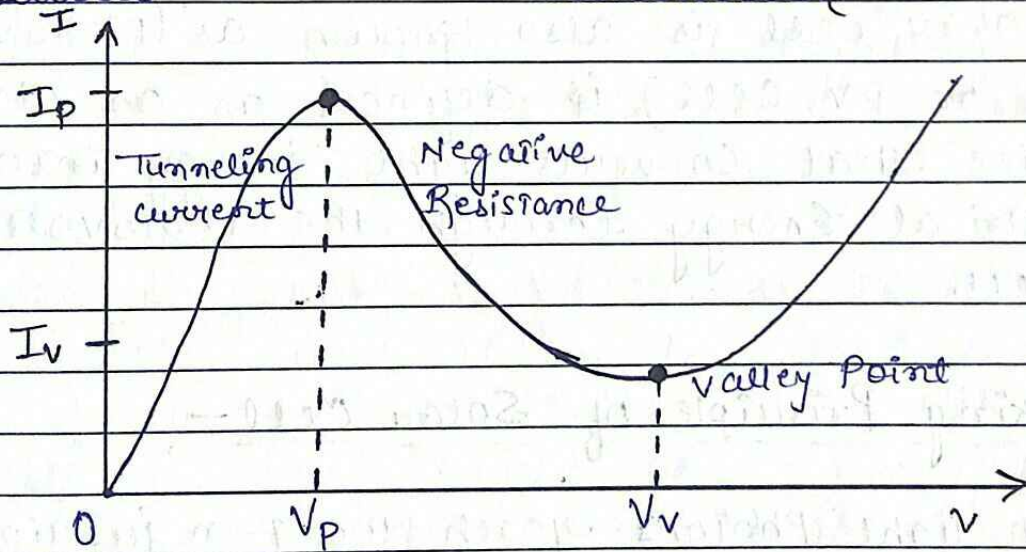
Working Principle - The working principle of Tunnel Diode is the same as a PN Junction diode, except high doping density alters its conductivity equal to a huge extent. In a PN

Junction Diode, the concentration of doping is 1 atom within 10^8 atoms. However, in Tunnel diode, the concentration of doping is 1 atom within 10^3 atoms. Because of the high impurity concentration of ions, the depletion layer width can be decreased & it turns into approx - 10^{-5} mm.

Circuit Diagram of Tunnel Diode -



Tunnel Diode VI Characteristics :-



ADVANTAGES :-

- The Switching Speed of this diode is high.
- long life
- This Diode handles with high frequencies.
- Fabrication is simple.
- less Noise & Cost.
- Power Dissipation is low
- operation is high Speed.
- Environmental Protection

DISADVANTAGE :-

- These Diodes are low - Power Devices.
- These are a bit expensive.
- It delivers less o/p voltage swing.

APPLICATIONS :-

- Tunnel Diode is used in different oscillators like Relaxation, Microwave, etc.
- It is used as a switching device with high speed.
- It is like a logic memory storage device.
- It is used in used in FM Receivers & Oscillator.
- It is used as a high frequency component.

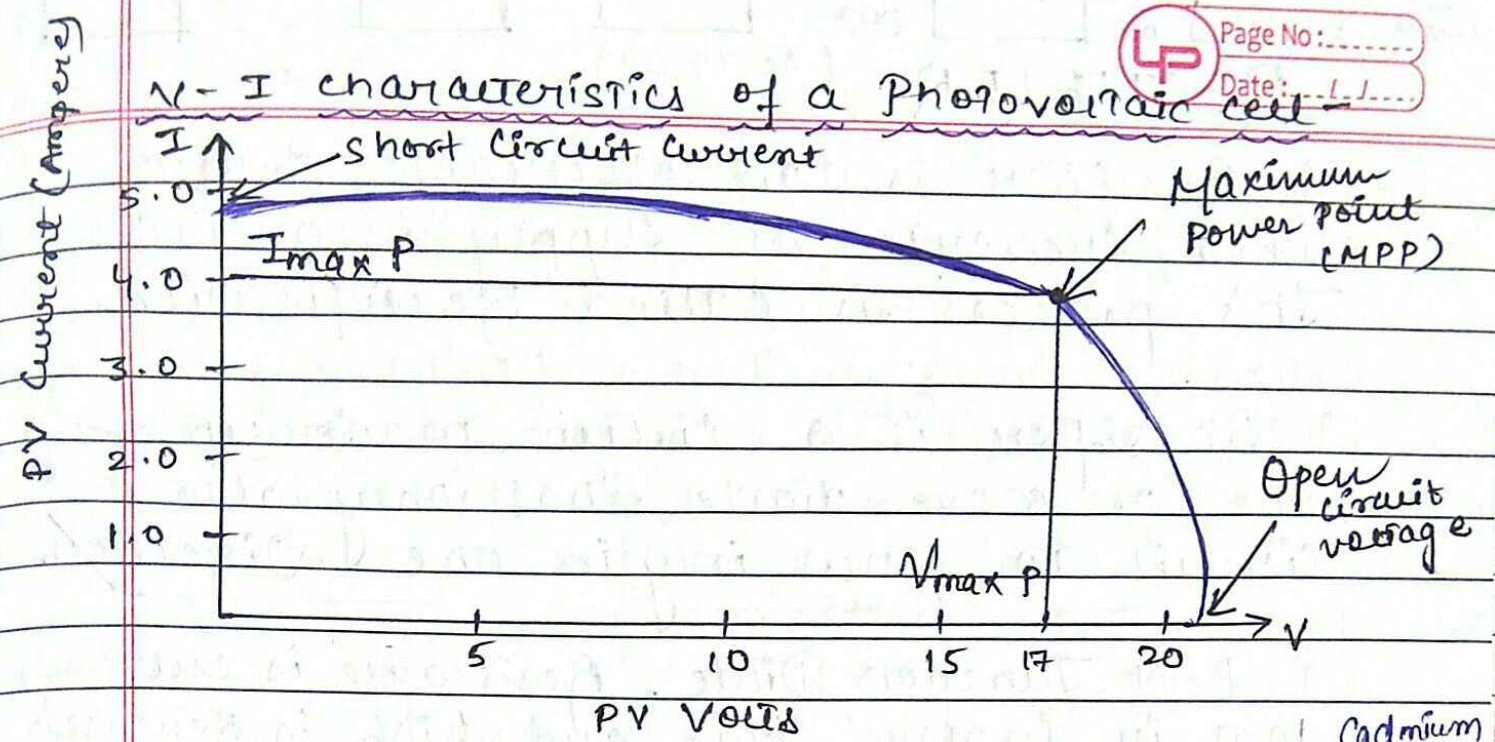
SOLAR CELL

A solar cell is also known as (Photovoltaic cell or PV cell) is defined as an electrical device that converts light energy into electrical energy through the photovoltaic effect.

Working Principle of Solar Cell -

When light photons reach the P-n junction through the thin P-Type layer, they supply enough energy to create multiple electron-hole pairs, initiating the conversion process. The incident light breaks the thermal equilibrium condition of the junction. The free electrons in the depletion region can quickly come to the n-type side of the junction.

V-I characteristics of a Photovoltaic cell



Cadmium
Telluride

Material used in solar cell - 1) Silicon 2) GaAs 3) CdTe

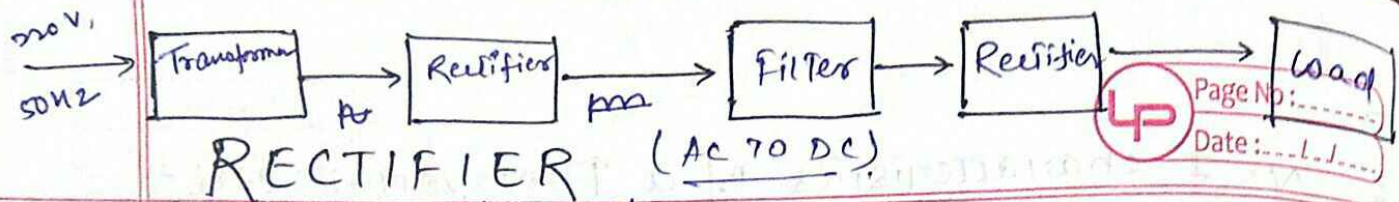
4) CuInSe₂
↳ (Copper Indium Selenide)

- ADVANTAGES :-
1. NO Pollution associated with it.
 2. It must last for a long time.
 3. NO maintainance cost.

- Disadvantages :-
1. It has high cost of installation.
 2. It has low efficiency.
 3. During cloudy day, the Energy cannot be Produced and also at night we will not get Solar Energy.

Uses :-

1. It may be used to charge batteries.
2. Used in light meters.
3. It is used to Power Calculators and wrist watches.
4. It Can be used in Spacecraft to Provide Electrical Energy.



A Rectifier is an electronic device which converts AC supply into DC. This process is called Rectification.

A Rectifier is a circuit made up of one or more diode that only allow current to flow only in one direction.

A P-N Junction Diode, Resistance is extremely low in forward bias and high in Reverse bias.

There are two types of Rectifier-

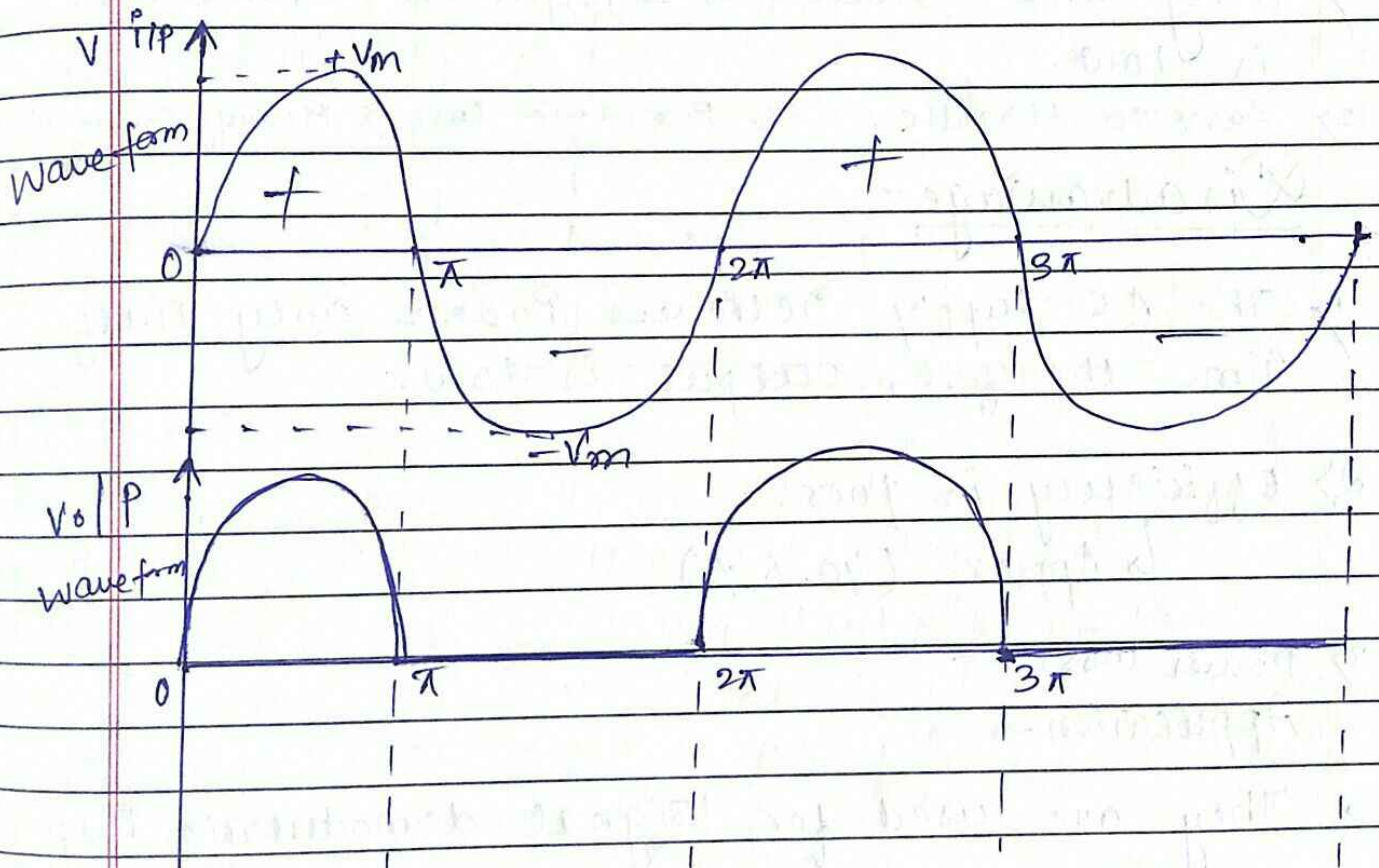
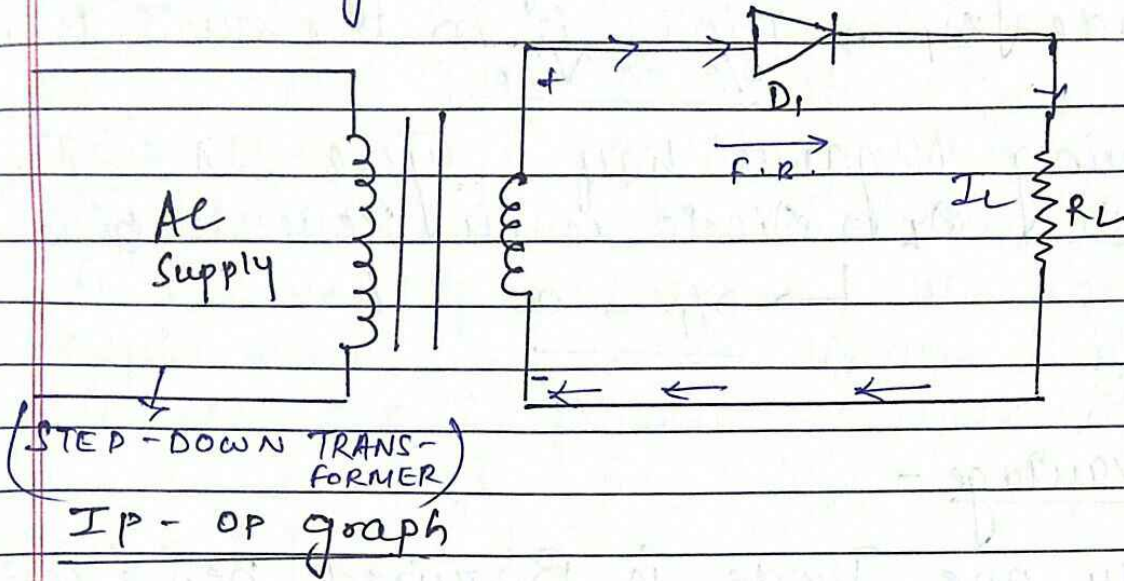
- 1) Half-wave Rectifier
- 2) Full-wave Rectifier

Full-wave Rectifier ^{further} Classified into two types-

- 1) Center-Tap Rectifier
- 2) Bridge Rectifier

* Half-wave Rectifier - In half-wave rectifier, the Rectifier conducts current only during Positive Half Cycle of AC Input Supply. During Negative Half cycle, no current is conducted hence

no voltage across the load.



During Positive Half - Cycle $0 - \pi$,
Diode $D_1 \rightarrow$ Diode is in Forward Bias.
 $\text{o/p} \rightarrow V_{\text{input}}$

During Negative half - Cycle, $\pi - 2\pi$,
Diode $D_2 \rightarrow$ Diode is in Reverse Bias.
 $\text{o/p} \rightarrow 0$

Advantage -

- 1) only one Diode is Required hence cost is low.
- 2) Easy to Handle.
- 3) Portable Cost & Manufacturing Cost.

Disadvantage -

- 1) The AC Supply Deliver Power only Half Time therefore, output is low.
- 2) Efficiency is Poor.
 $\rightarrow$ Approx - (40.6%)
- 3) Power loss

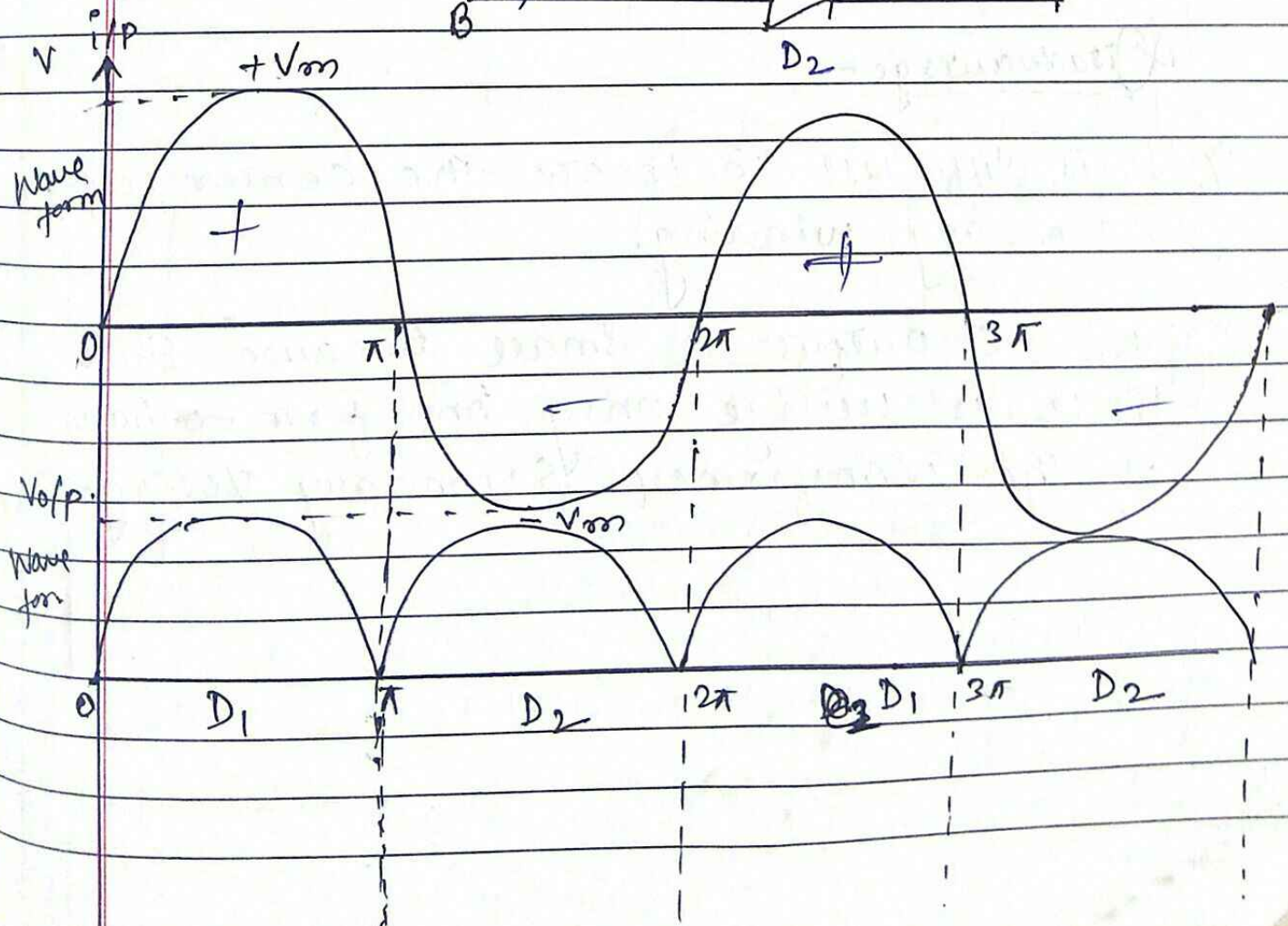
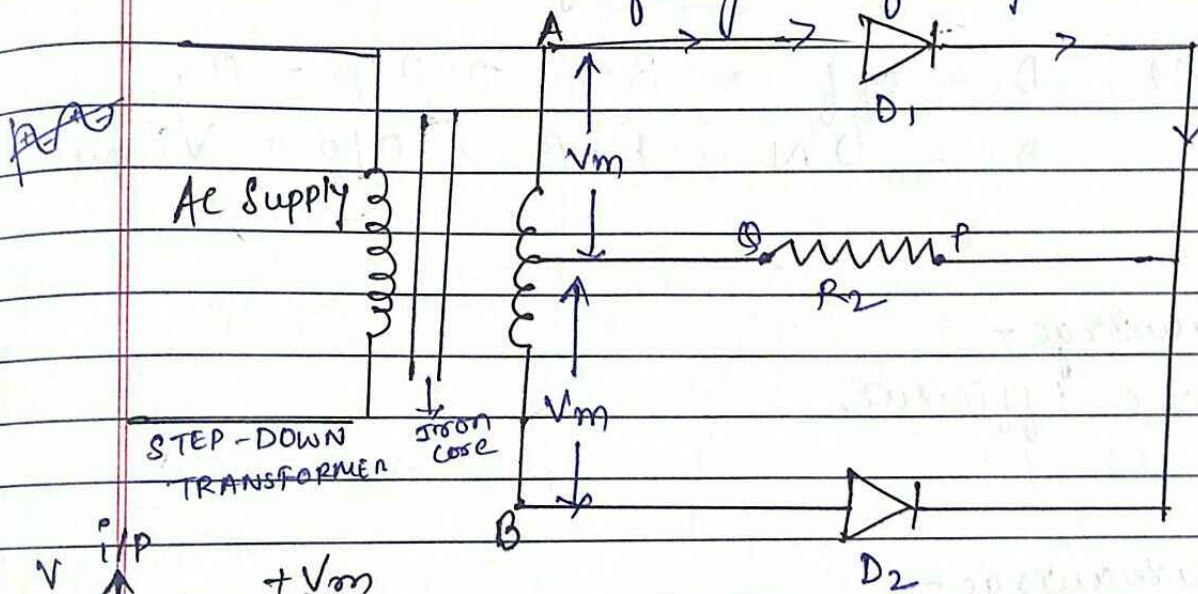
Application -

- They are used for signal demodulation Purpose.
- They are used for Rectification applications.
- They are used for signal Peak Applications.

Full-Wave Rectifier -

1) Centre-Tap Rectifier - During Po
It contains two diode D_1 and D_2 .

A Centre Tap Secondary winding AB is used with two diodes connected so that it uses one half cycle of input AC voltage.



Case 1 :- For +ve half cycle -

Diode $D_1 \rightarrow$ ON $\rightarrow$ Forward Bias, $O/P = V_{input}$

$D_2 \rightarrow$ off $\rightarrow$ Reverse Bias, $O/P \rightarrow 0$

Case 2 :- for -ve half cycle.

Diode, $D_1 \rightarrow$ off $\rightarrow$ R.B $\rightarrow$ $O/P = 0$.

$D_2 \rightarrow$ ON $\rightarrow$ F.B $\rightarrow$ $O/P \rightarrow V_{input}$

Advantage -

1) More Efficient.

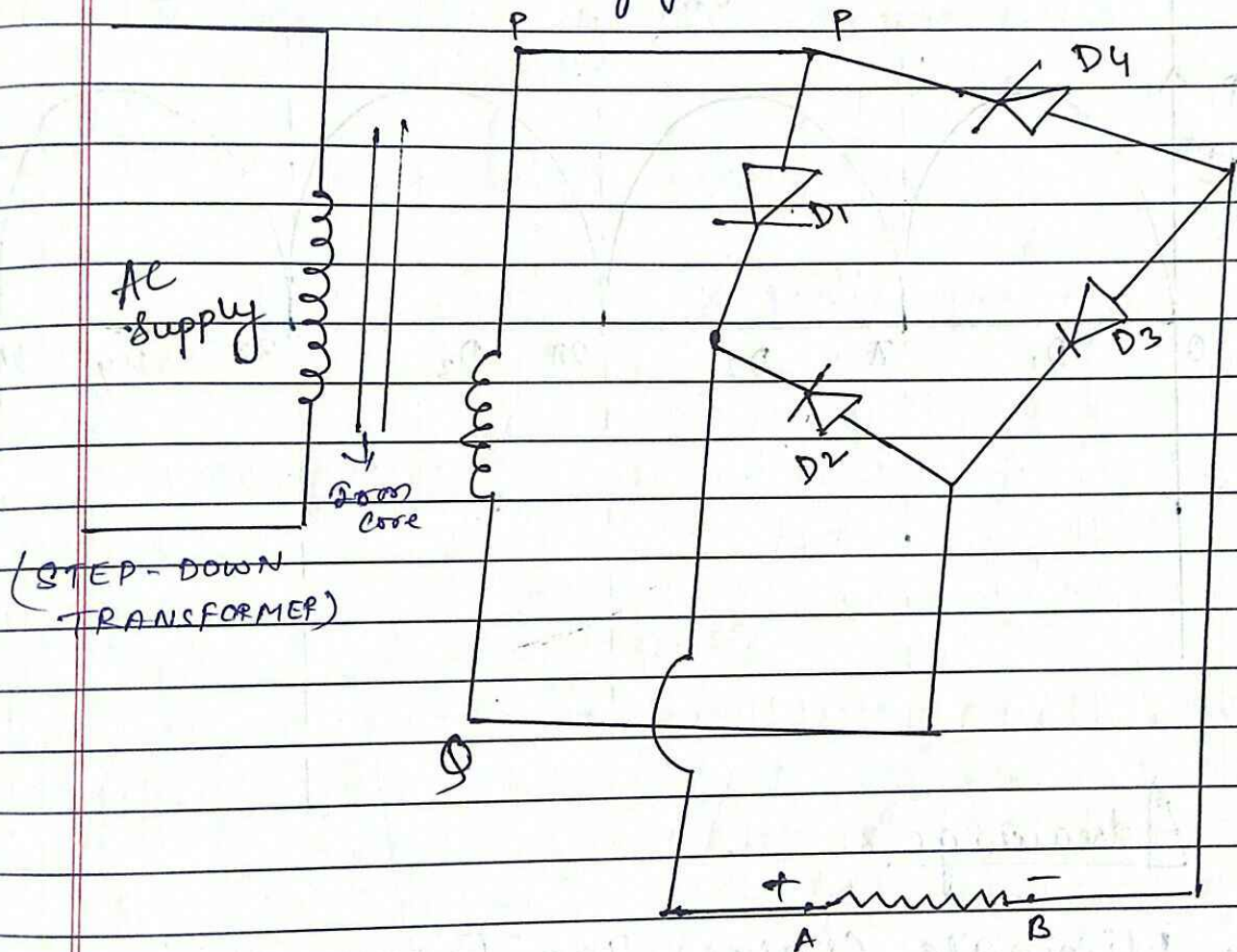
Disadvantage -

1) It is difficult To locate the Center Tap on Secondary winding.

2) The DC output is small because Each Diode is utilize only one ~~part~~ half of the transformer Secondary Voltage.

Full-Wave

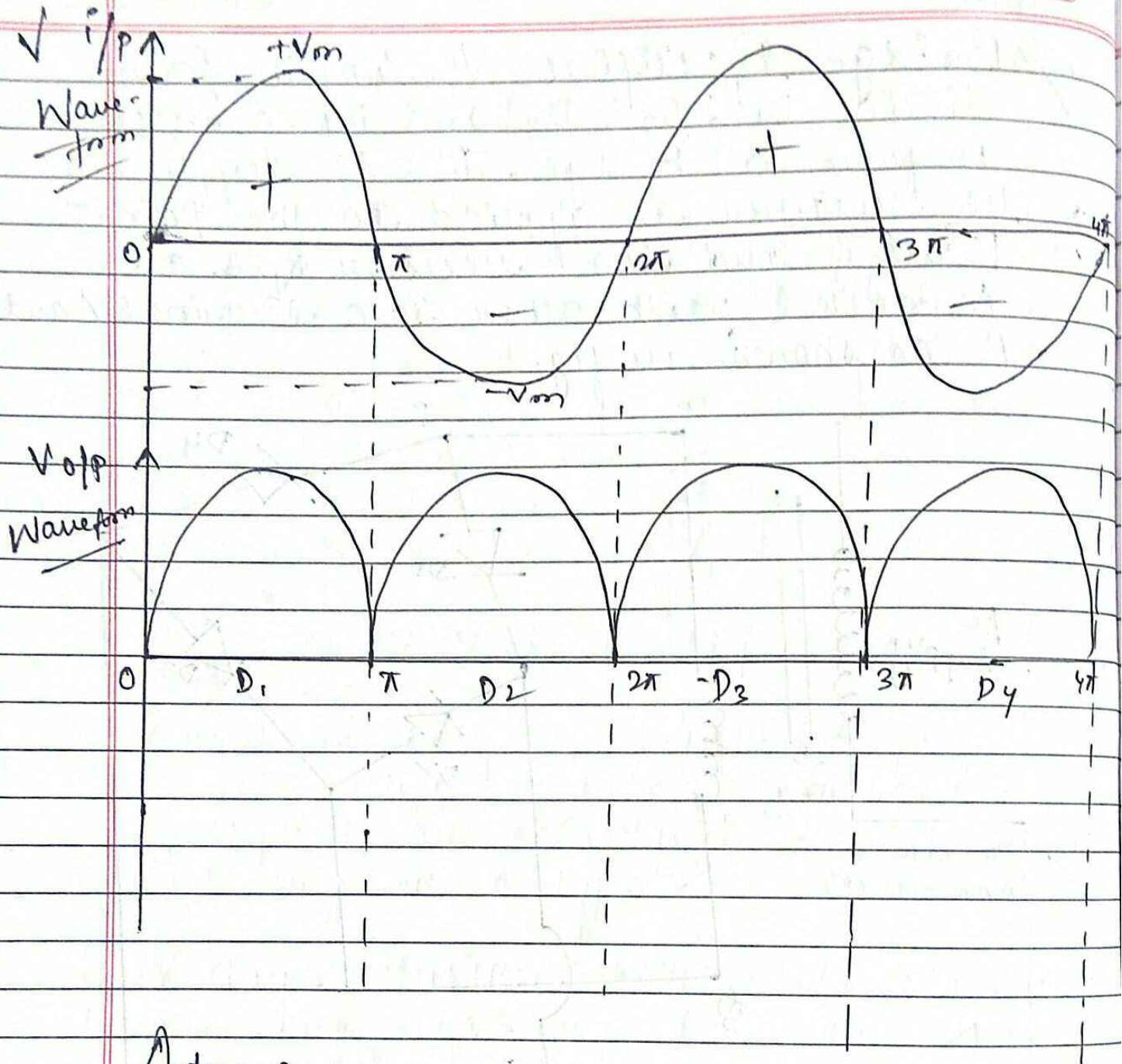
2. Bridge Rectifier - It contains four diodes D_1 , D_2 , D_3 and D_4 connected to form a bridge. The AC supply to be rectified is applied to the points P and Q and load resistor R_L is connected with other two terminals A and B as shown in fig.



Case 1: when P Terminal is +ve, & Q is -ve then, D_1 and D_3 are active and D_2 and D_4 are inactive.

Case 2: when P Terminal is -ve, Q is +ve then, D_2 & D_4 are active and D_1 and D_3 are inactive.

I/p - O/p Graph.



Advantage

- 1) Eliminate Centre-Tap Transformer
- 2) The Output is Twice that On Center Tap-Circuit for the Same Secondary Voltage.

Disadvantage

- ① Complex CK+ ② Charge & cost, is more.

Ripple factor of Rectifier

It is the measure of Purity of DC output of a rectifier. The Ripple factor measure the percentage of AC component in the Rectified output.

- Ripple factor high means A.C factor more.
- Ripple factor low means D.C factor more so,
- Ripple factor of Half wave rectifier $\rightarrow 1.21\%$.
- Ripple factor for full wave rectifier $\rightarrow 0.48\%$.

Efficiency of Rectifier (η):-

$$\eta = \frac{\text{O/p D.C Power}}{\text{I/p A.C Power}} \times 100$$

η for Half-Wave Rectifier $\rightarrow 40.6\%$

η for full-wave Rectifier $\rightarrow 81.2\%$

Voltage Multiplier

A voltage Multiplier is an electronic circuit consisting of Capacitors and Diode and is used to multiply or rise the voltage level of an AC signal.

The voltage multiplier receive an AC voltage of lower value, convert it into DC voltage and increases its voltage level.

Types of Voltage Multiplier -

Based on the ratio of the Input voltage to the output voltage, the Voltage Multiplier are classified into the following three types -

- 1) Voltage doubler → These circuit doubles the input voltage (Use two Diode and two Capacitor) or (2 D or 2 C)
- 2) Voltage Tripler → These circuit output is equal to the three times of the Maximum value of the input. (Use 3 Diode & 3 Capacitor)
- 3) Voltage Quadrupler → These circuit is a simply ~~sc~~ stacked combination of two voltage doubler circuit (Use 4 Diode & 4 Capacitor)

Voltage Doubler

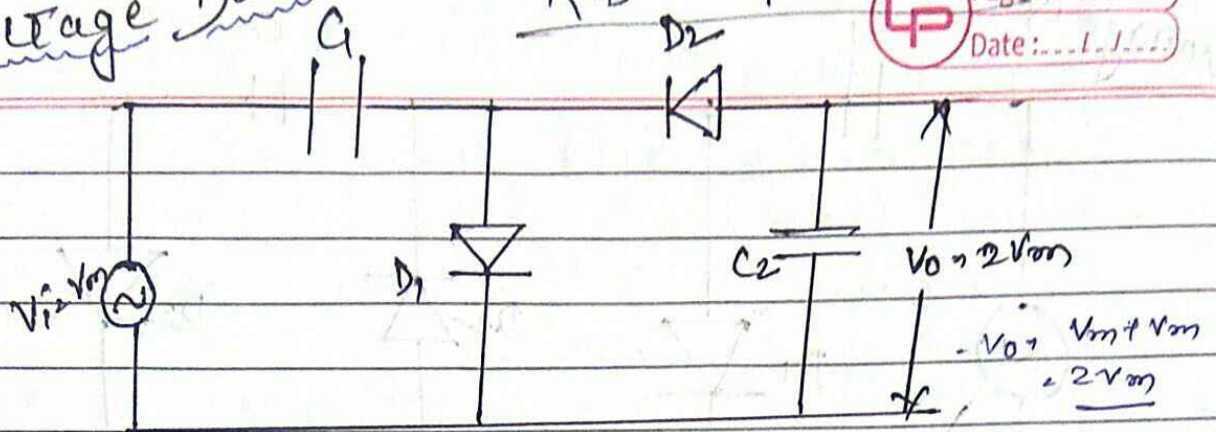
F.B $\rightarrow$ Short circuit

R.B $\rightarrow$ Open circuit

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Case 1:- for +ve half cycle -

$\rightarrow$ short circuit

Diode D_1 - ON (S.C)

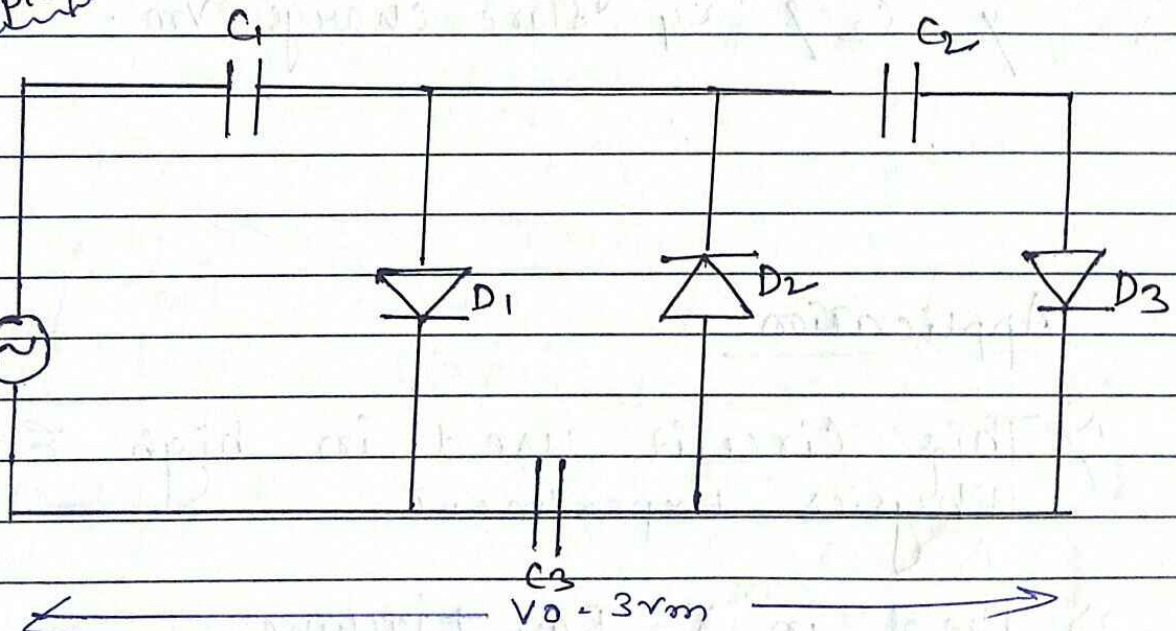
C_1 store charge ' V_m '

Case 2:- for -ve half cycle $\rightarrow$ short circuit

Diode D_2 - ON (S.C)

C_2 store charge ' V_m ' $\rightarrow$ So total O/P $\rightarrow V_m + V_m$
 $\rightarrow 2V_m$

Voltage Tripler



Case 1:- for +ve half cycle -

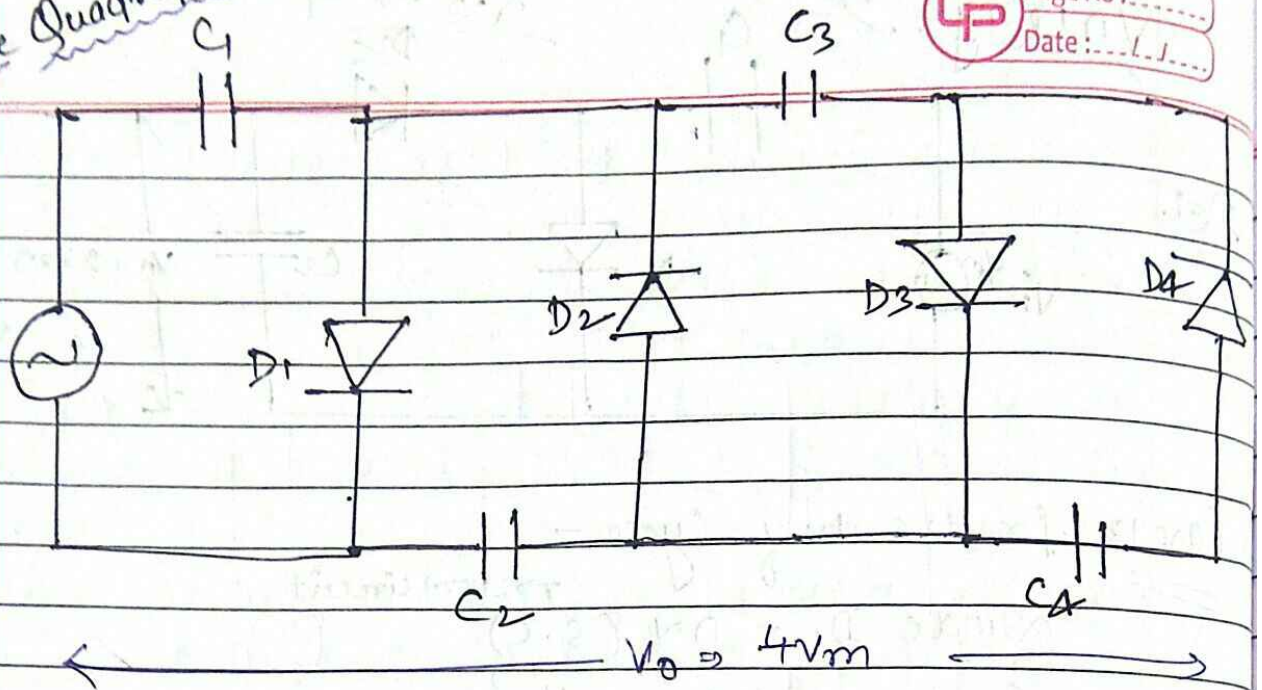
Diode - D_1 & D_3 - ON (Short Circuit).

C_1 and C_2 store charge ' V_m '.

Case 2:- for -ve half cycle - D_2 - ON (Short Circuit).

C_3 store charge ' V_m ' -

Voltage Quadrupler



Case 1: for +ve half cycle - D_1 & D_3 - ON (S.C.)
& ' C_1 ' & ' C_3 ' store charge ' V_m '.

Case 2: for -ve half cycle - D_2 & D_4 - ON (S.C.)
& ' C_2 ' & ' C_4 ' store charge ' V_m '.

Application -

- 1) This circuit used in high Energy Physics Experiment.
- 2) Used in X-Ray Machine
- 3) Used in Photo Multiplier Tube.
- 4) Used in Laser Printer and Photocopier machine.

Clipper Circuit (Wave Shaping) Page No. Date: .../.../...

Clipper Circuit are used to remove the part of a signal that is above or below some defined reference level.

One of the Simple Example of Clipper circuit is Half-Wave Rectifier that circuit basically let-off everything at the reference level zero and let only the Positive going or Negative going portion of the Input waveform.

Clipper Circuit also known as limiter, Amplitude Selector or Slicers.

Depending on the Orientation of ^{the} Diode the Positive and the Negative region of the input signal is clipped off.

Types of Clipper Circuit

1) Series Clipper Circuit

2) Parallel / shunt clipper circuit.

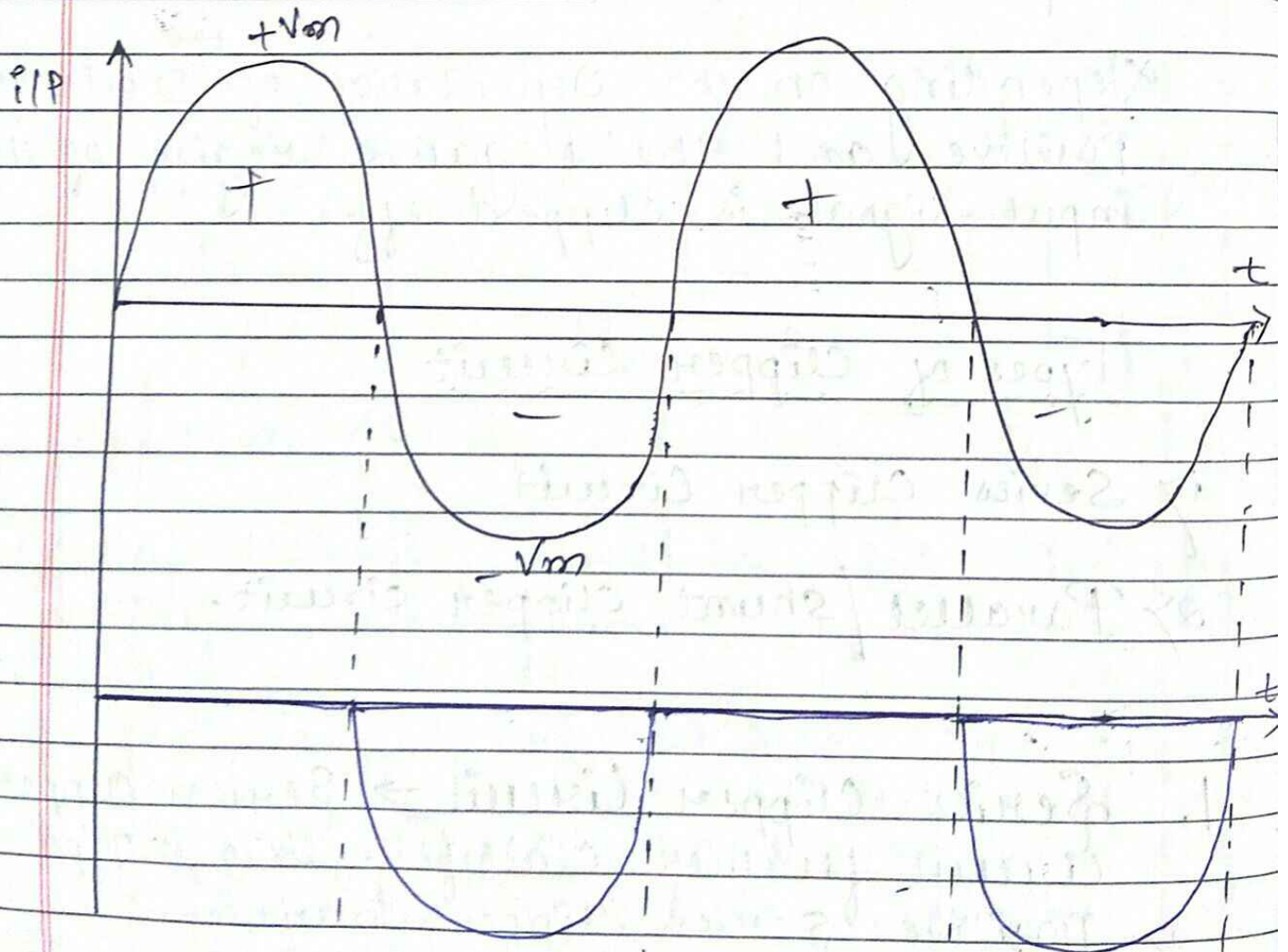
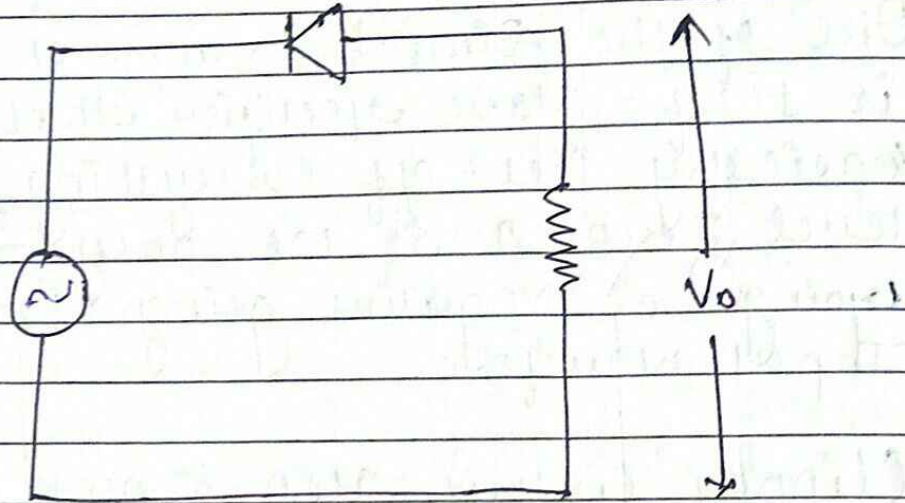
1. Series Clipper Circuit $\rightarrow$ Series Clipper circuit further classified into 2 Type -

a) Positive series clipper circuit -

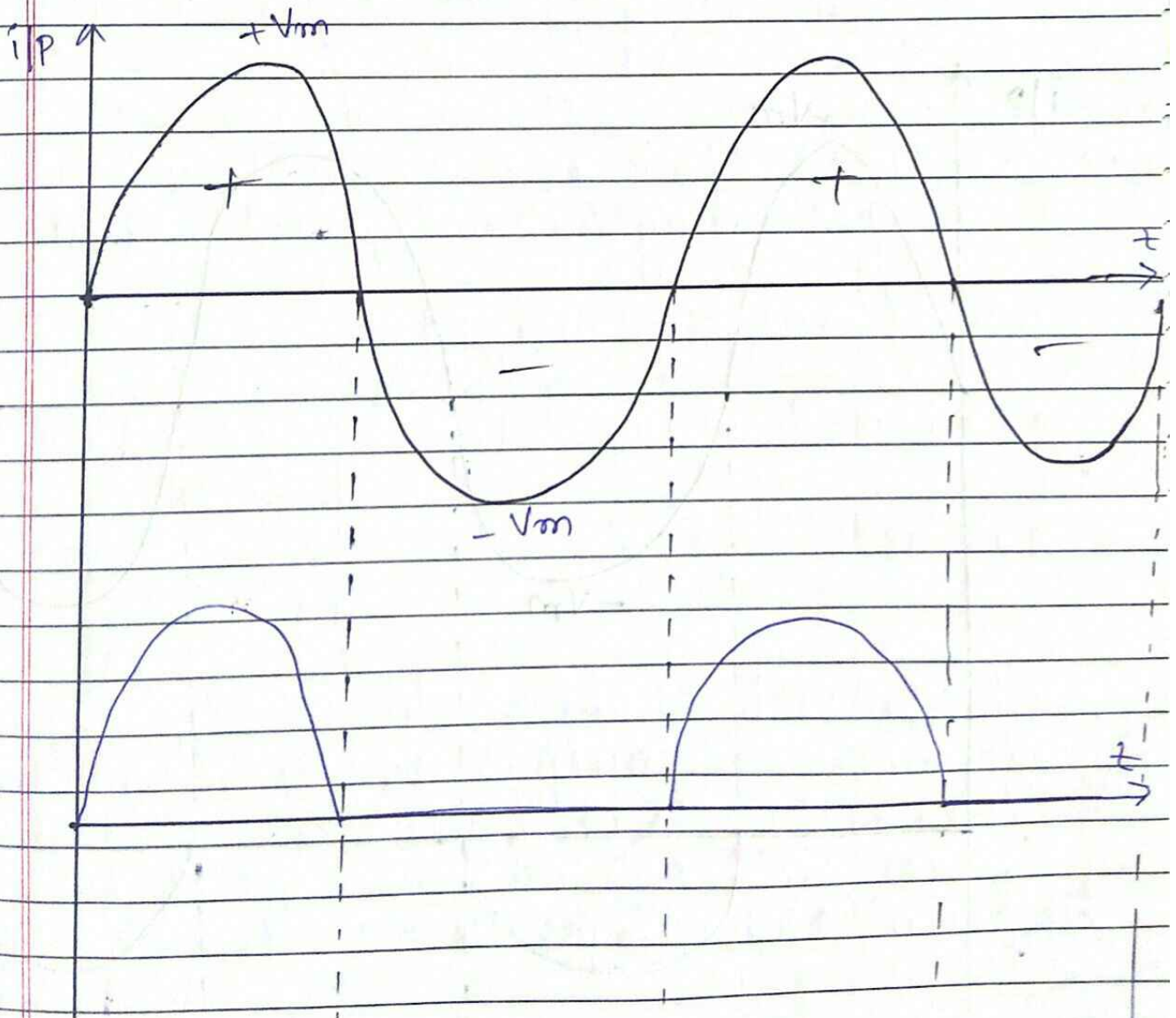
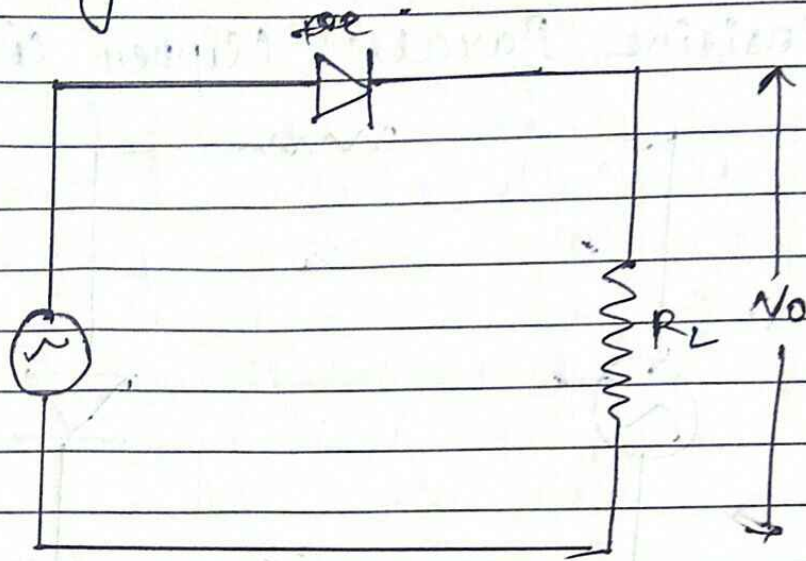
b) Negative series clipper circuit.

The Diode and the load resistor are connected in series of clipper circuit.

Positive
Series Clipper
Circuit

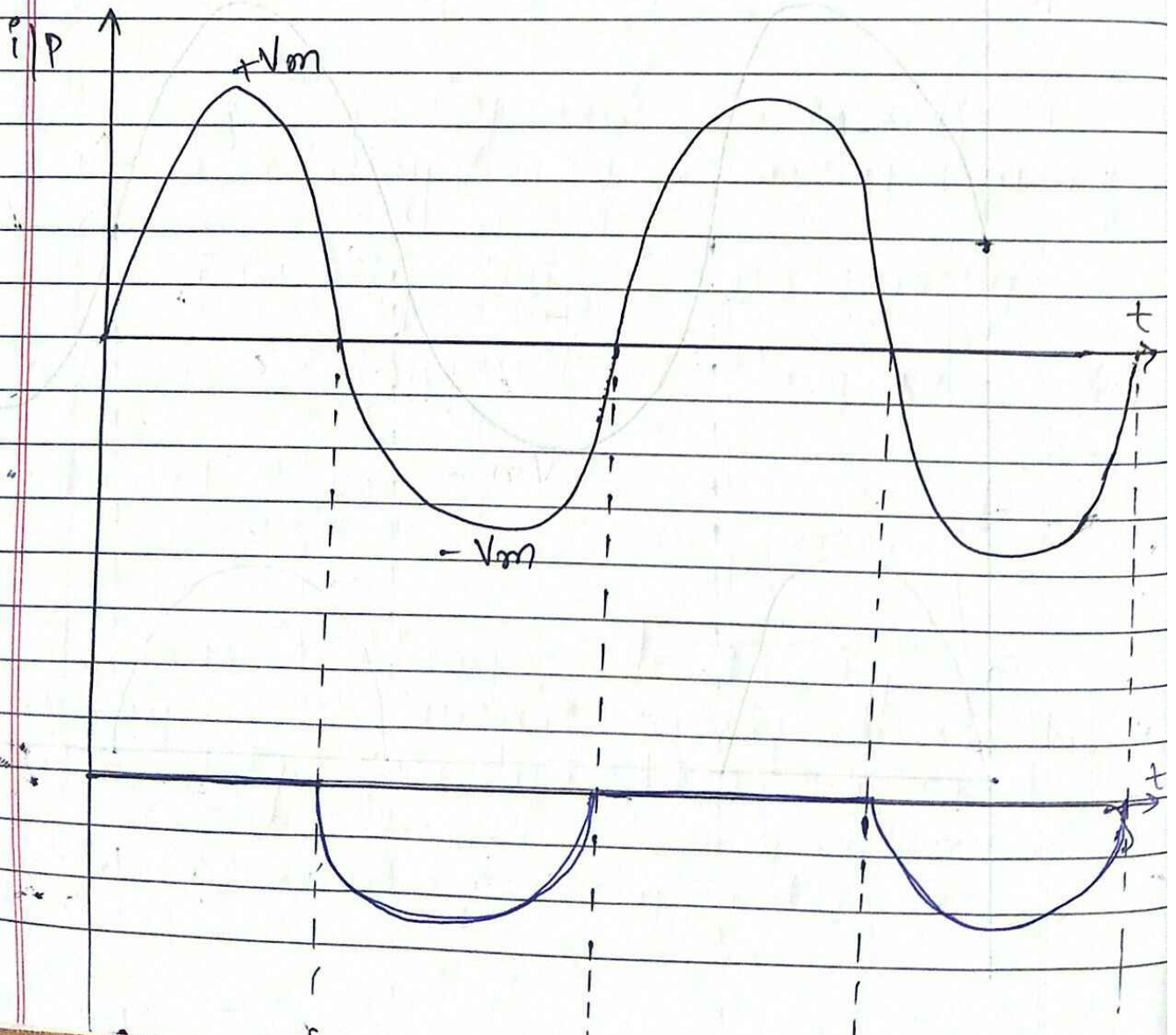
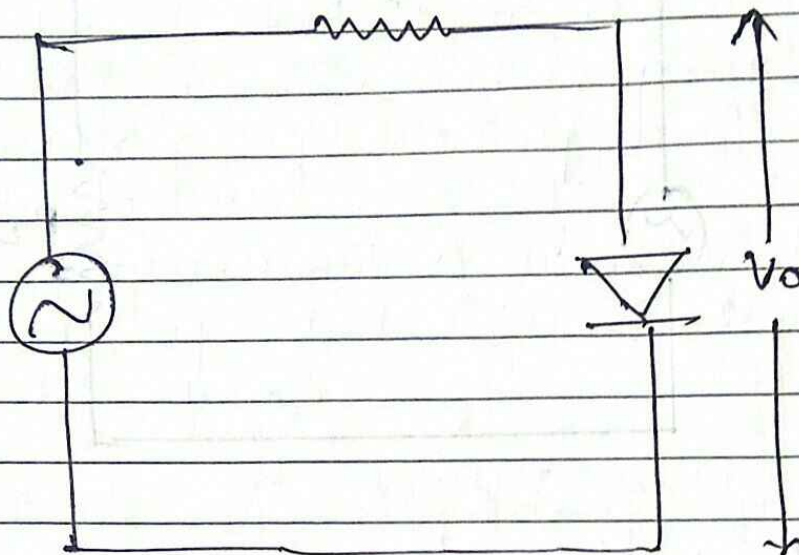


Series Negative Clipper -

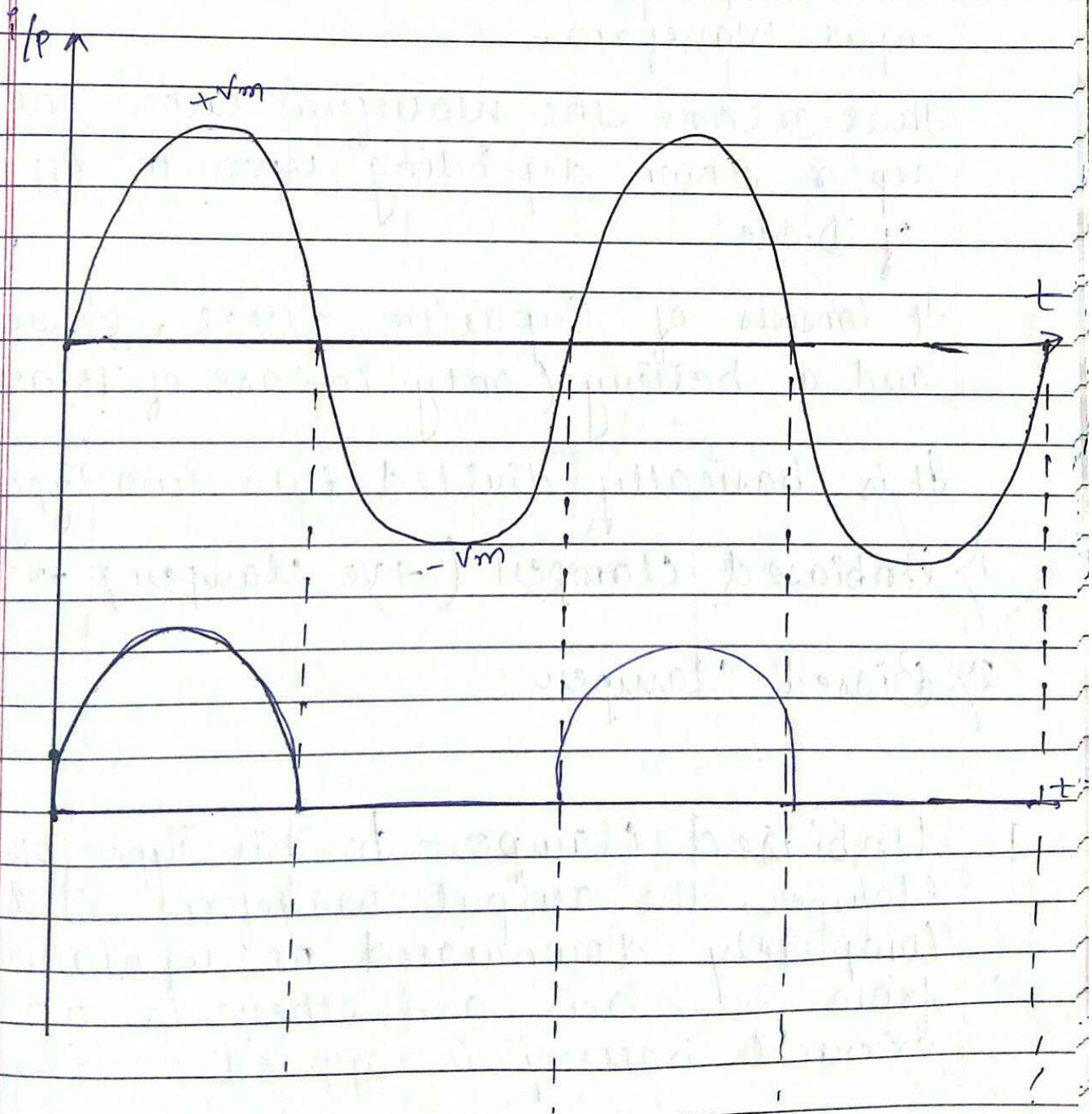
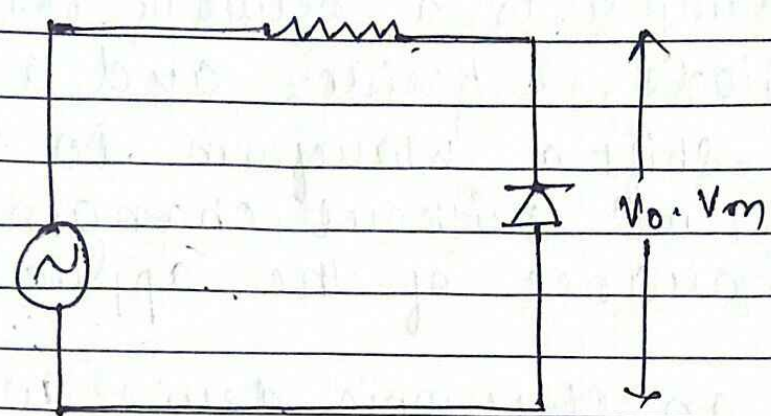


2) Parallel Clipper Circuit -

a) Positive Parallel Clipper Circuit.



b) Negative Parallel clipper Circuit -



Clamper -

A clamper is a network constructed of a diode, a resistor and a capacitor that shift a waveform to a different DC level without changing the appearance of the applied signal.

It is an electronic device which is used to 'clamp up' and 'clamp down', the input waveform.

That means the waveform either move up or down depending upon the Application of Diode.

It consists of Capacitor, Diode, Resistor and a battery (only in case of Biasing).

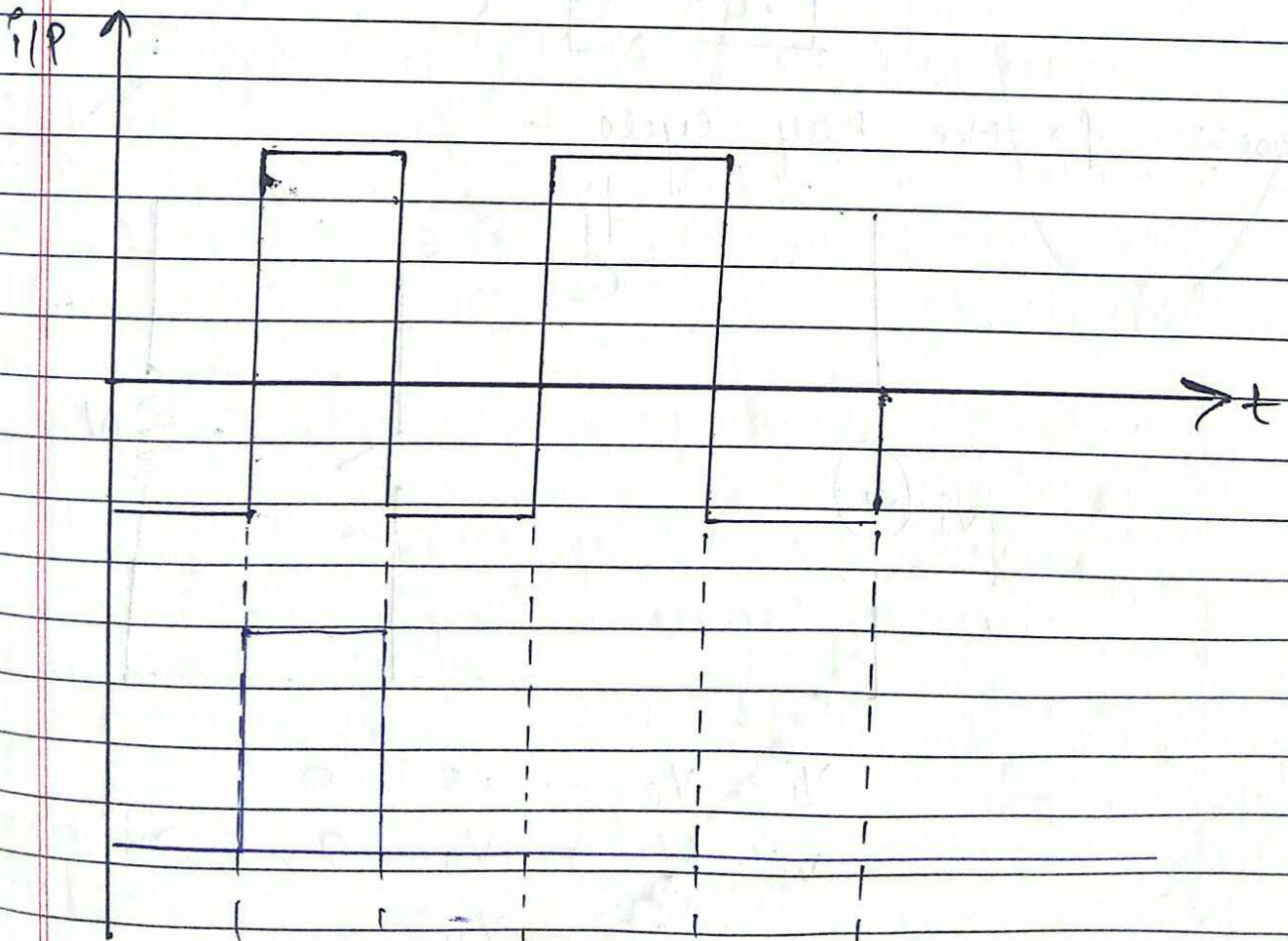
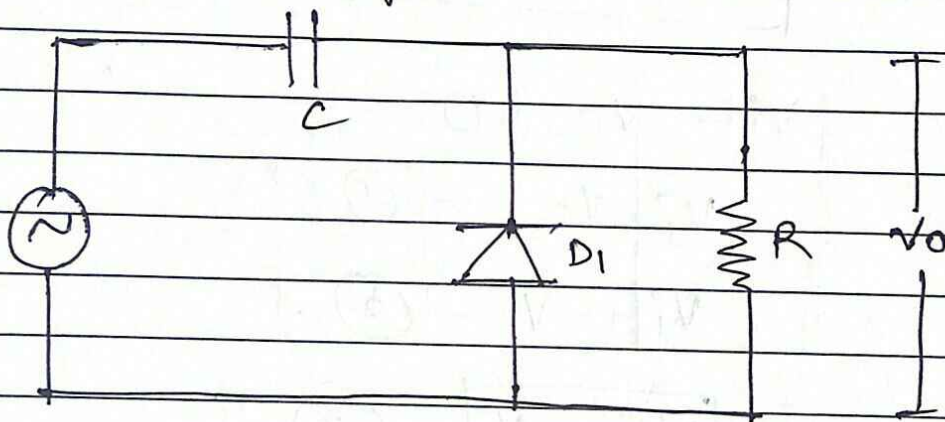
It is basically divided into two types -

- 1) Unbiased clamper (+ve clamper & -ve clamper)
- 2) Biased clamper

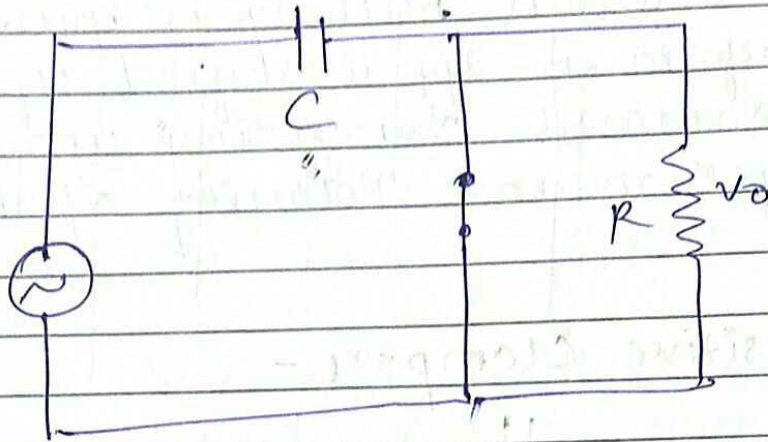
1. Unbiased Clamper - In this type of clamper, the output waveform is shifted completely downward or upward from zero axis and there is no external battery is applied.

2. Biased Clamper - In this type of Clamper, the external battery is applied and output waveform is shifted upward or downward depending upon the diode's direction and Polarity of the battery.

1. Positive clamper -



Case 1 - for -ve half cycle -



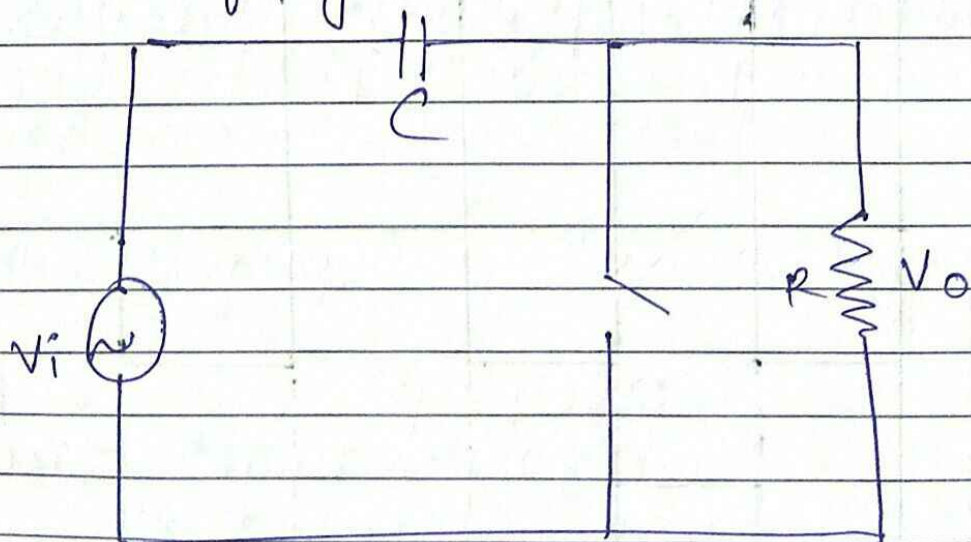
$$V_i - V_C = 0$$

$$V_i = V_C \quad \text{--- (1)}$$

$$V_i = V \quad \text{--- (2)}$$

$$V_C = V \quad \text{--- (3)}$$

Case 2 - for +ve half cycle -



$$V_i - V_C - IR = 0$$

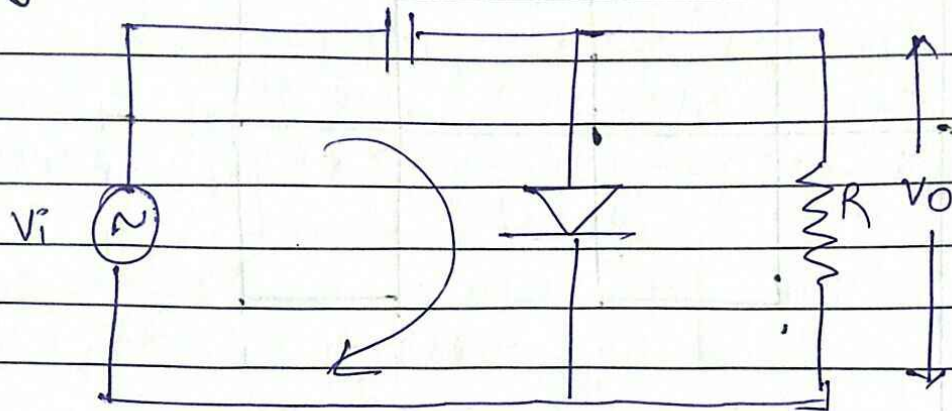
$$V_i - V_C - V_o = 0 \quad \rightarrow \text{from eq (1)}$$

$$V + V = V_o$$

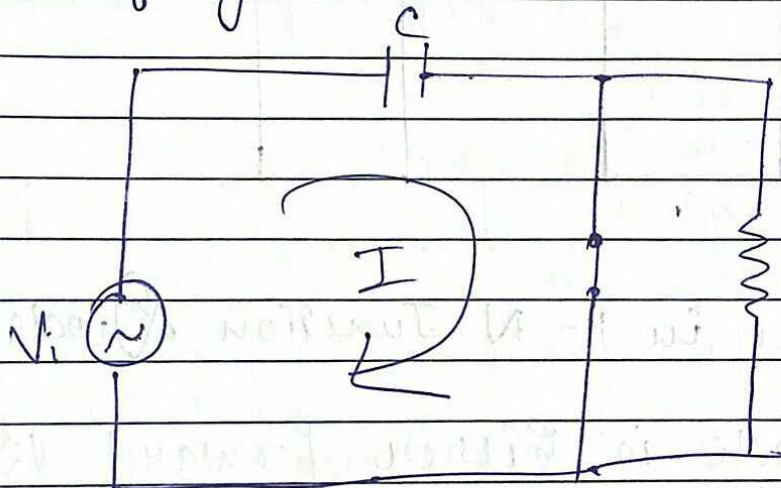
$$V_o = 2V$$

$$V_C = V$$

2. Negative Clamper -



1) For ^{+ve} half cycle - diode - F.B.

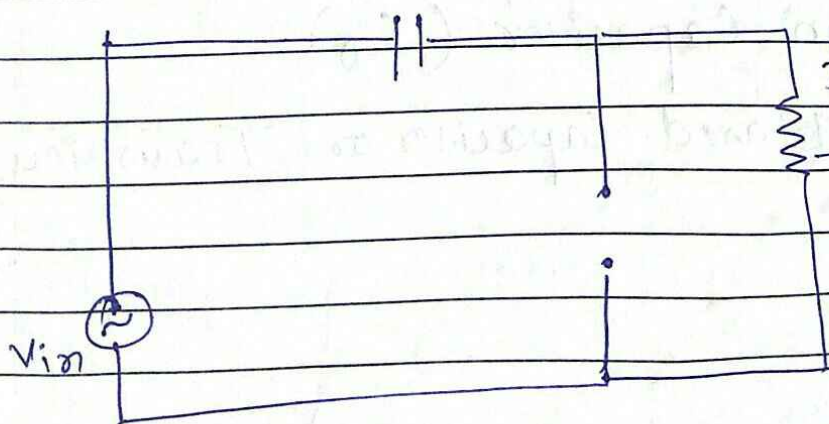


for +ve half cycle - diode - F.B

$$V_{in} - V_C = 0$$

$$V_{in} = V_C$$

$$V = V_C \quad \text{--- (1)}$$



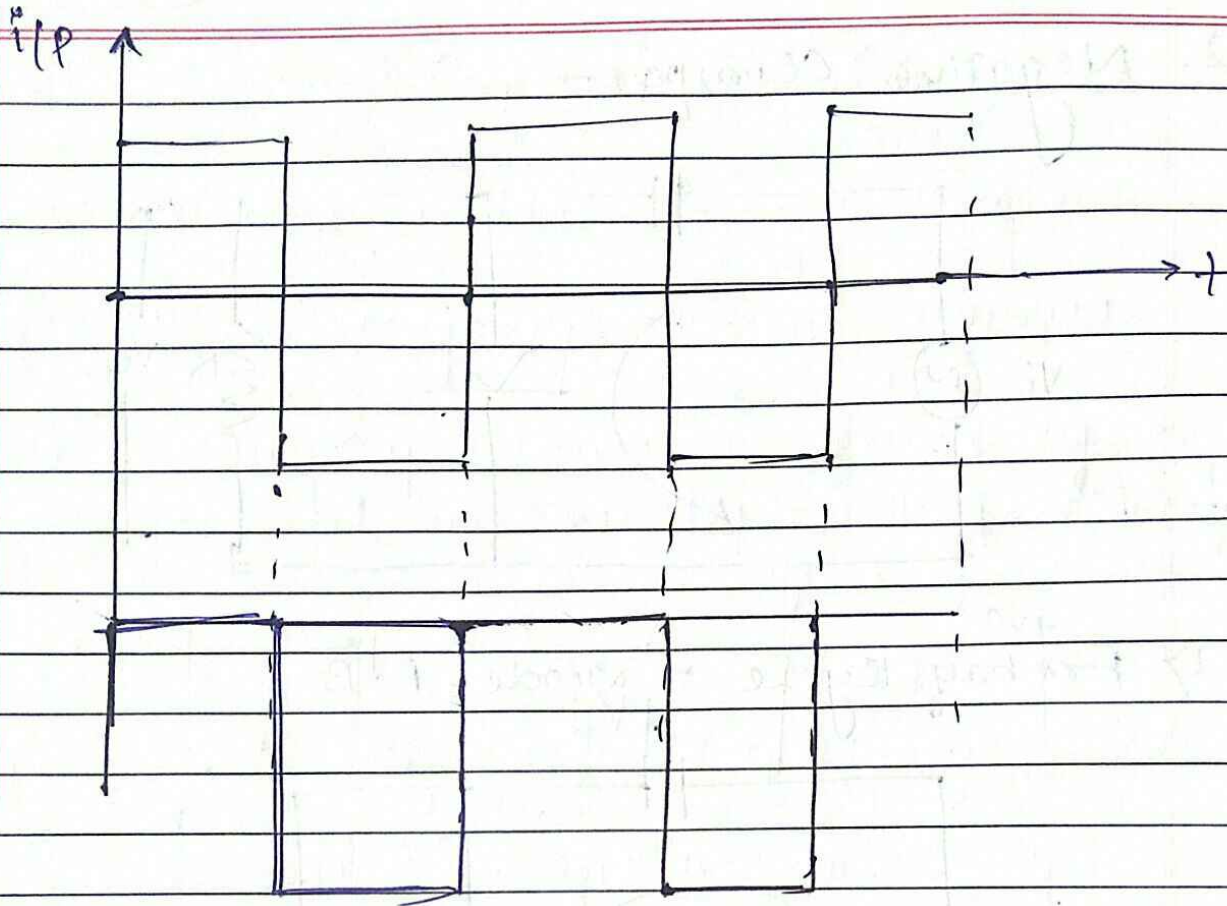
for -ve half cycle:-

$$-V_{in} - V_C - V_o = 0$$

$$-V - V = V_o$$

$$V_o = -2V$$

=====



Capacitance in P-N Junction Diode -

- when Diode is either forward Bias or Reverse Bias, Capacitance is formed in Diode.
- There are two types of Capacitance formed in Diode -
 - ① forward Biased Capacitor
 - ② Diffusion Capacitor (C_D)
 - ③ Reverse Biased Capacitor or Transition Capacitor (C_T)

Diffusion Capacitor (C_D) -

- It occurs under forward Bias. It stores minority charge carriers. C_D is directly proportional to forward current.

$$C_D \propto I_F$$

• ~~Step~~

$$C_D = \frac{\tau \cdot I_F}{\eta \cdot V_T}$$

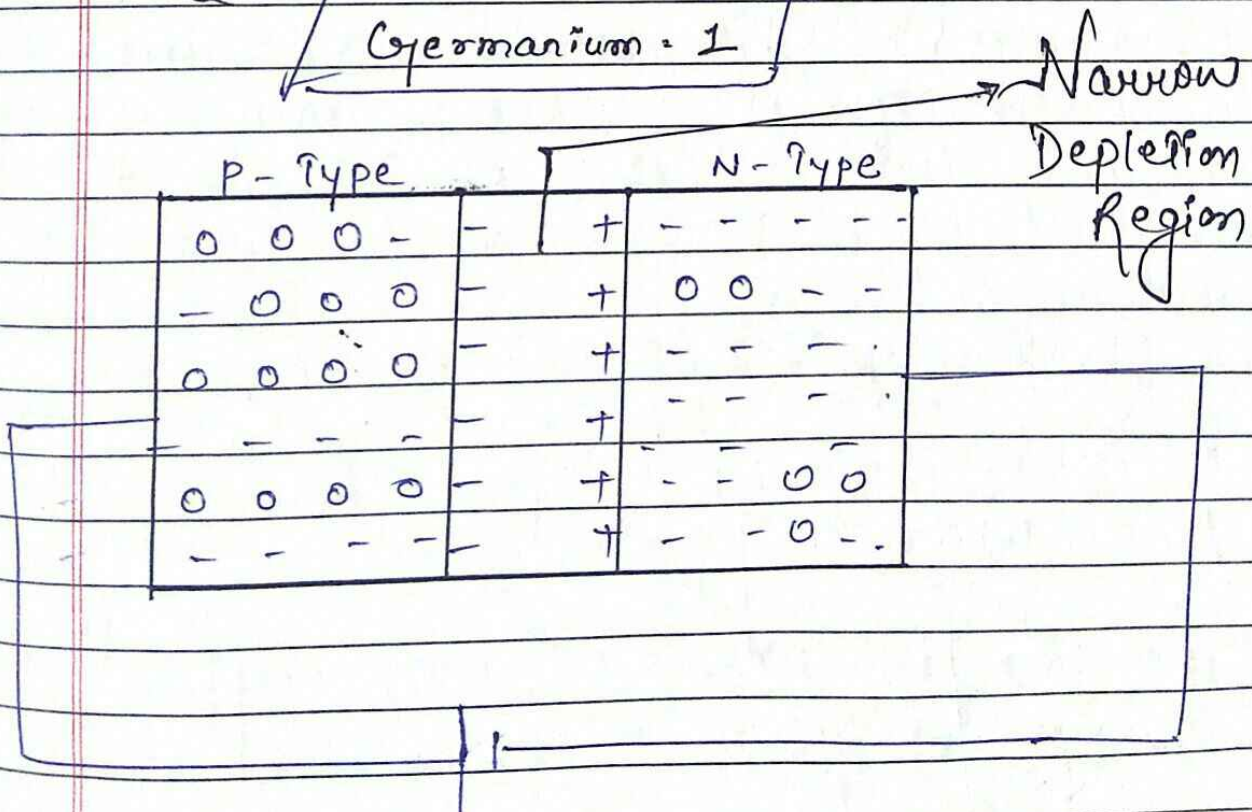
τ - Minority Carrier Lifetime

I_F - Forward Current

V_T - Thermal voltage

η - Constant.

~~200~~ $\text{Si} - 2$
 $\text{Germanium} - 1$



Transition Capacitance (C_T) -

- It occurs under Reverse Bias.
- It stores Majority charge carrier.
- Transition Capacitance is inversely proportional to width of Depletion layer.

$$C_T \Rightarrow \frac{\epsilon A}{W_d}$$

ϵ - Permittivity.

A - Area of Width

W_d - Width of Depletion layer.